HSM: Workload-Similarity Gating for Index-Maintenance Decisions, with Formal Bounds

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Abstract—Modern index advisors (BALANCE, MFIX, Indexer++, ShapeShifter, classical DTA/Dexter) answer *which* indexes to build, but invoke their costly $\Omega(N \log N)$ search at every workload window or on a fixed schedule. None addresses the orthogonal question of *when* the current workload has diverged enough from the last calibration window to warrant another invocation. We present HSM (Hierarchical Similarity Measurement), a training-free, schema- and DBMS-agnostic *decision layer* that sits in front of any such advisor and gates its invocation in $\Theta(N_{\text{pts}})$ time, with $N_{\text{pts}} \ll N$. HSM quantifies workload similarity along five orthogonal dimensions (query-template rank, volume, type distribution, attribute-set overlap, DWT-based temporal pattern) and exposes a single deploy-time threshold $\theta^*(N, Q)$. A formal theoretical core of six theorems and four lemmas establishes metric soundness, $\Theta(N_{\text{pts}})$ measurement complexity, economic gating optimality with a closed-form threshold, stratified-Hoeffding detector-quality bounds, a deployment speedup certificate, and a closed-form optimal window size; these results apply to every advisor satisfying the $\Omega(N \log N)$ premise (Table 4). On PostgreSQL 16 with TPC-H (SF 0.2–3.0, up to 18 M rows), HSM attains a discrimination ratio of 1.62 ($p < 0.001$), Youden $J \geq 0.96$ at every scale, and reduces advisor invocations by 80.8% versus always-on and 42.5% versus a periodic ($K = 3$) schedule, at a per-decision kernel cost of 1.09 ms ($< 0.1\%$ of query time). The closed-form optimal window size is empirically consistent across the 15× data-volume range under a cache-controlled measurement protocol. Four independent workloads (SDSS, JOB/IMDB, pgbench, burst) confirm generalizability (DR 1.09–2.01); cross-engine validation on MongoDB 7 demonstrates that the kernel transfers unmodified, with only the decision threshold requiring per-engine calibration ($J = 0.98$ at $\theta = 0.65$). HSM is not a physical index structure and not an index advisor: it is a lightweight *when-to-rebuild* layer whose $\Theta(N_{\text{pts}})$ decision cost is dominated by the $\Theta(N \log N)$ advisor work it gates.

Index Terms—Decision layer, index-maintenance timing, proactive index management, workload similarity, hierarchical similarity measurement, discrete wavelet transform, self-tuning databases.

I. Introduction

Database index performance is sensitive to workload characteristics. When the distribution of query types, access frequency, or temporal patterns changes substantially, an index configuration that was optimal for a previous workload window may become inefficient for the current one. Proactive index management—deciding whether to retain an existing index or rebuild it based on workload evolution—requires a reliable mechanism to determine how similar the current workload window is to the reference window under which the index was last calibrated. This paper addresses that prerequisite.

Prior work [34, 35] validated the HSM concept, demonstrating 77% classification accuracy and a 4,623× latency improvement over unoptimised baselines using a

coarse interval-overlap metric $S_i = |T_{\text{overlap}}|/|T_{\text{total}}|$. That metric cannot distinguish burst events, periodic sub-patterns, or multi-scale frequency structure, and those studies provided no formal theoretical analysis.

We introduce the HSM framework, making five contributions. **C1** A five-dimensional similarity measure with formal metric properties, capturing burst events, periodic oscillations, and long-term trends simultaneously. **C2** A theoretical core of six theorems and four lemmas covering wavelet pattern preservation, minimal-sufficient dimensions, economic gating optimality, stratified-Hoeffding detector-quality bounds, a deployment speedup certificate, and a closed-form optimal window size (Section IV). **C3** Access-attribute independence: HSM incurs no reconstruction cost when the accessed column set changes (Theorem 3). **C4** A closed-form threshold $\theta^*(N, Q)$ validated across SF 0.2–3 (Youden $J \geq 0.96$ at every scale). **C5** Cross-engine validation on MongoDB 7, confirming that the HSM kernel is DBMS-agnostic.

HSM is not a physical index structure but a proactive similarity measurement layer, yielding $\Theta(N_{\text{pts}})$ measurement cost dominated by the $\Theta(N \log N)$ reconstruction cost it gates. The experimental validation follows reproducibility principles established for database benchmarking [19].

II. Related Work

A. Workload Characterisation and Similarity Methods

Early work on workload compression [38] and automated index selection [8] treats workloads as static query sets via predicate overlap or template matching, ignoring temporal evolution and volume dynamics. BALANCE [44] groups workload chunks via self-supervised contrastive learning for index-selection instances, not temporal windows. MFIX [7] applies multi-fidelity Bayesian optimisation between advisor runs; Indexer++ [39] detects binary trend shifts via BERT embeddings and deep RL. None provides a continuous multi-dimensional similarity score with formal metric properties, nor do any address whether to invoke the advisor at a transition—all assume invocation is always warranted. HSM addresses this as an advisor-agnostic gating pre-filter. Recent ML-based index advisors such as BALANCE [44] and Indexer++ [39] employ contrastive learning and BERT embeddings with deep RL for end-to-end index selection. These approaches demonstrate strong performance on index *recommendation*; however, they target a complementary problem—determining *what* indexes to create—rather than *when* to invoke an advisor or rebuild an existing index. Their training pipelines assume GPU availability and labelled training corpora, assumptions that may not hold in resource-constrained or cross-engine deployments. HSM is designed as a lightweight, training-free gating layer that can be placed *in front of* any such advisor, reducing unnecessary invocations while preserving their recommendation quality. ShapeShifter [45] and the benchmark of [3] use similarity only as a binary trigger. Prior HSM work [34, 35] validated HSM empirically; the present paper adds formal proofs and scale sensitivity analysis.

Gap 2.1: To the best of our knowledge, no existing method jointly provides a continuous multi-dimensional workload similarity score with formal metric properties, a closed-form decision threshold, and a training-free deployment model for online indexing decisions (A9: 1.09 ms overhead, 0.25% of query exe-

cution time).

Deployed Index Advisors. Alongside academic prototypes, two open-source PostgreSQL advisors define the practical deployment state of automatic index recommendation: *Dexter* [21], which scans query logs and applies what-if analysis via HypoPG [18][1] to recommend B-tree indexes; and Supabase `index_advisor` [42], a PostgreSQL extension that drives the same HypoPG mechanism from an SQL interface. Both are widely deployed (Heroku/Instacart/Supabase production use), actively maintained through 2024, and have been adopted as baseline advisors in several recent industrial and academic studies. Open-source commercial derivatives include *pganalyze* Index Advisor and Huawei's *DBMind* [49]; the latter applies a reinforcement-learning policy in openGauss (a PostgreSQL fork) and therefore lies outside the vanilla-PostgreSQL protocol used in our experiments. Neither Dexter nor Supabase `index_advisor` addresses *when* to invoke advisor analysis—both assume the caller decides invocation externally—so HSM composes with them orthogonally by supplying the missing timing layer (Table 4).

B. Signal Processing for Time-Series Similarity

DWT [12] decomposes signals into multi-resolution components; Chan & Fu [6] applied this to time-series indexing. SAX [27] converts normalised series to discrete symbols with bounded error; a 2025 survey [32] reviews variants, including Quantile SAX [40] for non-Gaussian series. DTW [36] measures elastic similarity; FastDTW [37] approximates DTW in linear time, enabling $O(N)$ workload-window comparison. The three are complementary—multi-resolution decomposition (DWT), bounded-error symbolisation (SAX), and elastic alignment (FastDTW)—motivating the three-stage S_P pipeline (Section III).

Theoretical foundations. Metric space theory [14] formalises similarity properties. Information-theoretic bounds for symbolic representations draw on Cover & Thomas [11]. Wavelet approximation bounds are established in Mallat [28]. These classical results provide the mathematical scaffolding for Section IV.

Gap 2.2: No prior work provides formal guarantees for multi-scale workload similarity measurement—neither proving metric properties nor establishing parameter optimality bounds.

C. Database Optimisation and Access-Attribute Dependence

Self-tuning systems (AI-DBA [26], CDBTune [48], IA^2 [46], GPTuner [24]) all depend on workload characterisation as an upstream step; none provides a principled multi-dimensional similarity score with formal properties.

All physical index methods share *access-attribute dependence*: the structure is built for a fixed column set, and changing that set requires full reconstruction ($\Theta(N \log N)$). This holds from classical B-Trees [33] and Database Cracking [20], through first-generation learned indexes [23] (ALEX [16], PGM-Index [17], XIndex [43]), to multi-dimensional extensions (Flood [30], Tsunami [15], WaZI [31]). Table 3 (Section IV) provides a formal comparison.

Gap 2.3: No existing framework—from classical B-Tree and cracking to learned indexes (PGM, ALEX, XIndex) and workload-aware spatial partitioning (WaZI)—formally characterises access-attribute independence as a structural property, provides a continuous multi-dimensional similarity score across temporal workload windows, or quantifies the asymptotic cost advantage of proactive measurement over reactive rebuild.

D. NoSQL and Document Store Indexing

Cattell's taxonomy [5] established the SQL/NoSQL divide as a data-model and consistency tradeoff, but did not address workload similarity measurement. On the schema side, NoSE [29] formulates NoSQL physical design as an integer program over a conceptual data model; however, it targets Cassandra's wide-column model and does not consider temporal workload drift. Klettke et al. [22] address schema evolution and data migration in document stores at scale, highlighting that agile NoSQL development produces frequent schema versions—a setting where proactive drift detection is especially valuable. De Espona Pernas et al. [13] propose query-only automatic index suggestion for MongoDB aggregation pipelines, the closest NoSQL-specific index advisor to date; their method does not model temporal windows or workload similarity. On the SQL side, industrial index managers such as AIM [47] and workload-reduction techniques like Wred [4] remain relational-only. Stonebraker and Pavlo's recent retrospective [41] observes that relational and document models increasingly coexist in modern deployments, yet no cross-engine workload similarity framework has been proposed.

Gap 2.4: No existing index management framework—relational or NoSQL—provides a DBMS-agnostic workload similarity measure validated across both SQL and document-store engines with formal metric guarantees.

These four gaps map directly to contributions C1–C5 (Section I): Gaps 2.1–2.2 $\rightarrow$ C1–C2 (framework + theorems), Gap 2.3 $\rightarrow$ C3–C5 (access-attribute independence, threshold calibration, scale validation via Theorems 3–5), Gap 2.4 $\rightarrow$ cross-engine MongoDB validation (Section 5.7).

III. The HSM Framework

HSM measures the similarity between two workload windows W_A and W_B, each spanning a fixed interval of duration T and containing N_{pts} time-stamped query events. The framework operates in four phases: (1) preprocessing, (2) five-dimensional similarity computation, (3) score aggregation, and (4) index decision.

A. Workload Preprocessing

Each window W is reduced to: a k-element query-template frequency vector $\mathbf{f} \in \mathbb{R}^k$ (one entry per distinct template, extracted by the DBMS-specific extractor; Section III-B); a QPS time series $q(t)$ at 1-second resolution giving N_{pts} points[2]; relation set $\mathcal{T}$ and column set $\mathcal{C}$ forming the access-attribute schema $A(W) = (\mathcal{T}, \mathcal{C})$; and total volume $\text{QPS} = \sum_i f_i / T$.

B. Five Similarity Dimensions

Ratio Similarity S_R measures the relative-frequency distribution of query templates via Spearman rank correlation on the normalised query-frequency rank vectors $\mathbf{r}$ (one entry per distinct template, derived from $\mathbf{f}$):

$$S_R(W_A, W_B) = 1 - \frac{\arccos\big(\rho_s(\mathbf{r}_A, \mathbf{r}_B)\big)}{\pi}, \qquad (1)$$

[1] HypoPG creates hypothetical indexes whose presence can be tested through the PostgreSQL query planner without the $\Theta(N \log N)$ cost of physical construction.

[2] The implementation normalises $q(t)$ by $\max_t q(t)$ before the wavelet decomposition (`hsm_v2_kernel.py`: `sp_v2`). This makes the bound constant $M = 1$ in Theorem 1, so the cited tight bound $\varepsilon \leq 0.071$ at $L = 3$, $N_{\text{pts}} = 100$ holds without further scaling.

where $\rho_s(\cdot)$ is the Spearman rank correlation. Since $\rho_s \in [-1, 1]$, we have $\arccos(\rho_s) \in [0, \pi]$, so $S_R \in [0, 1]$. Importantly, $d_R = 1 - S_R = \arccos(\rho_s)/\pi$ is the normalised *correlation angle* and satisfies the triangle inequality analytically [14]. $S_R = 1$ iff both windows share identical query-frequency rank order. **Volume Similarity** S_V uses log-scale to handle order-of-magnitude differences:

$$S_V(W_A, W_B) = \exp(-|\ln Q_A - \ln Q_B|) = \frac{\min(Q_A, Q_B)}{\max(Q_A, Q_B)}. \tag{2}$$

Type Similarity S_T compares query-template distributions via angular distance on unit-normalised frequency vectors. Let $\mathcal{Q} = \{q_1, \ldots, q_k\}$ be the set of k distinct operation categories observed in the monitoring period (e.g. SQL query templates {Q1, Q3, ...} for relational DBMS; pipeline categories {`find`, `aggregate`, `insert`, `update`, `delete`} for document DBMS). Define $\mathbf{v} \in \mathbb{R}^k$ as the normalised frequency count over $\mathcal{Q}$ within window W:

$$S_T(W_A, W_B) = 1 - \frac{2}{\pi} \arccos\big(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B\big), \tag{3}$$

where $\hat{\mathbf{v}}$ denotes the ℓ_2-normalised unit vector. Frequency counts are non-negative, so $\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B \in [0, 1]$, giving $S_T \in [0, 1]$. The dissimilarity $d_T = (2/\pi) \arccos(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B)$ is the *angular distance* (geodesic) on $\mathbb{S}^{k-1}$, satisfying the triangle inequality for any $k \geq 1$ [14]. This DBMS-agnostic formulation allows $\mathcal{Q}$ to be defined by a *DBMS-specific extractor* (Section III-B), making S_T applicable across heterogeneous database paradigms.

Access Similarity S_A measures schema-level overlap as an equal-weight average of two Jaccard scores over table-level and column-level access sets:

$$S_A = \tfrac{1}{2} J(\mathcal{T}_A, \mathcal{T}_B) + \tfrac{1}{2} J(\mathcal{C}_A, \mathcal{C}_B), \quad J(X, Y) = \frac{|X \cap Y|}{|X \cup Y|}. \tag{4}$$

Equal weighting ensures that structural access and fine-grained attribute access carry equal diagnostic weight.

Pattern Similarity S_P captures multi-scale temporal behaviour through a three-stage pipeline. *Stage 1 (DWT).* Daubechies-4 (db4) at level $L = 3$ decomposes $q(t)$ into approximation coefficients $\mathbf{cA}_3$ and detail coefficients $\mathbf{cD}_3$, $\mathbf{cD}_2$, $\mathbf{cD}_1$ at three scales. *Stage 2 (SAX).* Each sub-band coefficient vector is encoded via SAX with alphabet size $\alpha = 4$. *Stage 3 (FastDTW).* FastDTW with Sakoe–Chiba radius $r = 3$ computes the alignment distance; the per-band score is normalised to $[0, 1]$: $\text{band_score} = 1 - \text{dist}/(\text{len}(s) \cdot (\alpha - 1))$.

$$S_P = w_3^A \text{sc}(\mathbf{cA}_3) + w_3^D \text{sc}(\mathbf{cD}_3) + w_2^D \text{sc}(\mathbf{cD}_2) + w_1^D \text{sc}(\mathbf{cD}_1), \tag{5}$$

with $w_3^A = 0.40$, $w_3^D = w_2^D = w_1^D = 0.20$ (sum = 1), where $\text{sc}(\cdot)$ denotes the per-band normalised score. The level-3 approximation band $\mathbf{cA}_3$ receives greater weight (0.40) to capture long-range trend; the three detail bands carry equal weight (0.20 each) to detect burst and periodic events at multiple scales.[3]

[3]To enforce the metric properties of Lemma 1 the implementation (`hsm_v2_kernel.py`: `_band_score`) (i) returns 1.0 when the two SAX encodings are byte-identical (P2 self-distance) and (ii) symmetrises the FastDTW alignment by averaging $\frac{1}{2}[\text{fastdtw}(s_a, s_b) + \text{fastdtw}(s_b, s_a)]$ (P3 symmetry). FastDTW is an approximation of full DTW, so without these two corrections P2 and P3 hold only up to the FastDTW radius; the implementation therefore satisfies Lemma 1 more tightly than the reference statement.

C. DBMS-Specific Feature Extraction

The five HSM dimensions define a *universal similarity kernel* whose inputs are DBMS-agnostic WorkloadWindow objects. A lightweight *extractor* bridges the gap between raw DBMS traces and these objects, encapsulating all DBMS-specific semantics:

Relational extractor (PostgreSQL/MySQL). $\mathcal{Q}$ = set of SQL query templates identified by the query planner's normalised statement text. $\mathcal{A}$ = {table.column} sets derived from the query access map. Volume Q = query count per window.

Document extractor (MongoDB). $\mathcal{Q}$ = operation categories {`find`, `aggregate`, `insert`, `update`, `delete`}, further resolved to pipeline-stage signatures for aggregation queries. $\mathcal{A}$ = {`collection.field_path`} accessed per operation (respecting nested-document paths). Volume Q = operation count per window.

The similarity kernel (formulas for S_R–S_P) is identical for both extractors; only the vocabulary $\mathcal{Q}$ and attribute set $\mathcal{A}$ change. DBMS-specific default weights are set to balanced priors and validated by sensitivity analysis on representative traces (Section 3.4, Experiment A10b).

D. HSM Score Aggregation

$$S_{\text{HSM}}(W_A, W_B) = \sum_{i=1}^{5} w_i S_i(W_A, W_B), \tag{6}$$

with $\sum_i w_i = 1$, $w_i \geq 0.05$ (all dimensions necessary, Theorem 2). Default weights are chosen as balanced priors, then validated by sensitivity analysis (A10b): DR changes by at most 5.5% under ± 0.05 single-weight perturbation. PostgreSQL defaults: $w_R = 0.25$, $w_V = w_T = w_A = 0.20$, $w_P = 0.15$. Table 1 summarises the five dimensions.

Table 1: Summary of HSM similarity dimensions. Each dimension maps a pair of workload windows to $[0, 1]$; the composite S_{HSM} is a convex combination (Eq. 6). Cost is per window-pair evaluation.

Dim.	Formula	Range	Cost	Captures
S_R	Spearman+arccos	$[0, 1]$	$O(N_{\text{pts}})$	Query-type freq. ranks
S_V	Log-exp	$(0, 1]$	$O(N_{\text{pts}})$	Query volume
S_T	Angular dist.	$[0, 1]$	$O(k)$	Query-template dist.
S_A	Avg. Jaccard	$[0, 1]$	$O(\|A\|)$	Attribute overlap
S_P	DWT+SAX+DTW	$[0, 1]$	$O(N_{\text{pts}})$	Temporal pattern
S_{HSM}	Convex sum	$[0, 1]$	$O(N_{\text{pts}})$	All five dims

E. Index Decision Rule

Definition 1 (Decision Threshold θ)**.** *The threshold $\theta \in (0, 1)$ is the minimum HSM score below which the expected cost of retaining the current index exceeds the expected cost of rebuilding it. Given $h = S_{\text{HSM}}(W_{\text{cur}}, W_{\text{ref}})$: if $h \geq \theta^*(N, Q)$ then* RETAIN*; if $h < \theta^*(N, Q)$ then* REBUILD*. The closed-form derivation of θ^* is in Theorem 3.*

F. Algorithm 2: HSM Index Diagnosis

When the gate fires ($h < \theta^*$), the operator needs to know *which* dimension is responsible for the regime shift in order to target a single index repair instead of a full rebuild. Algorithm 2 implements that diagnosis as a one-shot dimension-wise contribution analysis that is $O(1)$ on top of an HSM evaluation.

Algorithm 2 (HSM Index Diagnosis).
Input: per-dimension scores $\{S_d\}_{d \in \mathcal{D}}$ with $\mathcal{D}$ =

$\{R, V, T, A, P\}$ and weights $\{w_d\}$ from W_0.
Output: blamed dimension d^* and contribution vector $\mathbf{c}$.

1. For each $d \in \mathcal{D}$, compute the shortfall contribution $c_d = w_d\,(1 - S_d)$.
2. Return $d^* = \arg\max_d c_d$ and the full vector $\mathbf{c} = (c_R, c_V, c_T, c_A, c_P)$.
3. *Counterfactual repair check:* compute $\Delta S_{\mathrm{HSM}} = w_{d^*}(1 - S_{d^*})$, the closed-form gain to HSM if dimension d^* were perfectly repaired.

The decomposition is exact because $1 - S_{\mathrm{HSM}} = \sum_d w_d(1 - S_d)$, so $c_d/(1 - S_{\mathrm{HSM}})$ is the share of the gap attributable to dimension d. Algorithm 2 thus turns a scalar gate decision into an attributable diagnosis without any extra DBMS state and without any tuning.

Robustness model. In deployment the operator measures noisy estimates $\hat{S}_d = \mathrm{clip}(S_d + \varepsilon_d,\, 0,\, 1)$ with $\varepsilon_d \sim \mathcal{N}(0, \sigma^2)$ i.i.d. across dimensions. Algorithm 2 then chooses $\hat{d}^* = \arg\max_d w_d(1 - \hat{S}_d)$, which can disagree with the clean argmax when two dimensions have similar contributions. The operator's repair, however, acts on the *true* state and is scored on the clean S_d: gap-closed $= (S_{\mathrm{HSM}}(\mathrm{repair}(S, \hat{d}^*)) - S_{\mathrm{HSM}}(S))/(1 - S_{\mathrm{HSM}}(S))$. We report, for each noise level $\sigma \in \{0.00, 0.05, 0.10, 0.20\}$, both the agreement rate $\Pr[\hat{d}^* = d^*]$ and the mean gap-closed over $K = 100$ Monte Carlo trials per cross-phase pair. The clean ($\sigma = 0$) case recovers the exact decomposition above; the noisy cases characterise how much measurement budget the operator must invest to keep Alg. 2 strictly above the random-repair baseline.

G. Access-Attribute Dependence

Definition 2 (Access-Attribute-Dependent Method)**.** *A method M for index management is* access-attribute-dependent *if: (a) M constructs or maintains a physical data structure optimised for a fixed access-attribute set $A^* = (\mathcal{T}^*, \mathcal{C}^*)$; (b) when the observed access-attribute set $A(W_c) \neq A^*$, M must reconstruct its physical structure to remain optimal; and (c) that reconstruction incurs $\Theta(N \log N)$ cost.*

This definition applies universally—B-Trees, cracking, and all learned indexes satisfy it. HSM is not access-attribute-dependent: as a measurement function reading pre-aggregated window statistics, it remains logically unimpaired by any change in the physical access column, consistent with Codd's principle of physical data independence [10]. Theorem 3 formalises this property.

IV. Theoretical Analysis

We establish six core theorems supported by four lemmas. Lemmas 1–4 collect the foundational properties that justify the construction (metric axioms, parameter selection, complexity bounds, and a wavelet sublemma). Theorem 1 quantifies the wavelet approximation error of S_P. Theorem 2 proves the five HSM dimensions are minimal and sufficient. Theorem 3 unifies the economic break-even analysis with HSM's access-attribute independence under a single cost model. Theorem 4 establishes a stratified-Hoeffding concentration bound on the *intrinsic* detector quality $(\mathrm{TPR}, \mathrm{TNR})$ obtained by reframing HSM as a binary classifier of phase boundaries. Theorem 5 transports that band into a deployment-time certificate on the asymptotic speedup, parameterised by the operator's phase-mix prior π_w. Theorem 6 closes the framework with a closed-form optimal window size N^*_{pts} that minimises total steady-state cost.

A. Supporting Lemmas

Lemma 1 (HSM Metric Properties)**.** *$\forall\, W_i, W_j, W_k \in \mathcal{W}$: $(P1)$ $S_{\mathrm{HSM}}(W_i, W_j) \in [0, 1]$ (non-negativity and boundedness); $(P2)$ $S_{\mathrm{HSM}}(W_i, W_i) = 1$ (identity of indiscernibles); $(P3)$ $S_{\mathrm{HSM}}(W_i, W_j) = S_{\mathrm{HSM}}(W_j, W_i)$ (symmetry); $(P4)$ $d_{\mathrm{HSM}}(W_i, W_k) \leq d_{\mathrm{HSM}}(W_i, W_j) + d_{\mathrm{HSM}}(W_j, W_k)$ where $d_{\mathrm{HSM}} = 1 - S_{\mathrm{HSM}}$ (triangle inequality).*

Proof sketch. Each $d_i = 1 - S_i$ is an analytic metric; convex combination preserves boundedness (P1), identity (P2), symmetry (P3), and triangle inequality (P4). Full proof: Supplementary §I.

Lemma 2 (DWT Parameter Selection)**.** *Let $k = 4$ be the number of canonical workload states (volume regime $\times$ query-type dominance). The minimum SAX alphabet size that guarantees correct state discrimination under the maximum-entropy model with the two-sided burst extension satisfies $\alpha^* = 2k = 8$; under Quantile-SAX breakpoints, $P(\mathrm{misclass.}) < 1/\alpha = 1/8$. (Implementation uses $\alpha = 4$, the half-precision setting empirically validated in Section V.)*

Lemma 3 (Wavelet Coefficient Bound)**.** *For any signal $q(t)$ on $[0, N_{\mathrm{pts}})$ with $\sup|q(t)| \leq M$ and its L^2 norm $\|q\|_2 \leq M\sqrt{N_{\mathrm{pts}}}$, the level-$L$ db4 DWT approximation $\tilde{q}$ satisfies $\|q - \tilde{q}\|_2 \leq C^L_{\mathrm{db4}}\, M\sqrt{N_{\mathrm{pts}}}$, with $C_{\mathrm{db4}} = \sqrt{2}/2 \approx 0.707$ [28].*

Lemma 4 (Tight Linear Complexity)**.** *$T_{\mathrm{HSM}}(N_{\mathrm{pts}}, N) = \Theta(N_{\mathrm{pts}})$ in both upper and lower bound, independent of database cardinality N; space complexity is $\Theta(N_{\mathrm{pts}})$. The lower bound holds for any function $\mathrm{sim} : \mathcal{W} \times \mathcal{W} \to [0, 1]$ obeying P1–P4: such a function must inspect $\Omega(N_{\mathrm{pts}})$ elements of each input window, since otherwise an adversary can perturb an unread element to flip the value of sim while preserving P1–P4.*

Proof sketch. Upper: each dimension $O(N_{\mathrm{pts}})$ with no page access; total $\Theta(N_{\mathrm{pts}})$. Lower: adversary argument. Full proof: Supplementary §III.

B. Theorem 1: Wavelet Pattern Preservation

Theorem 1 (Wavelet Pattern Preservation)**.** *Let $q(t)$ satisfy $\sup|q(t)| \leq M$. The level-L db4 DWT used by S_P introduces pattern-similarity error bounded by*

$$\varepsilon(M, N_{\mathrm{pts}}, L) \;=\; \frac{2\, C^L_{\mathrm{db4}}\, M}{\sqrt{N_{\mathrm{pts}}}}, \qquad C_{\mathrm{db4}} = \frac{\sqrt{2}}{2} \approx 0.707, \tag{7}$$

which is strictly decreasing in both L and N_{pts}. Equation (7) is a closed-form expression in operator-measurable quantities.

Proof sketch. Follows from Lemma 3 and the normalised FastDTW score sensitivity. Full proof: Supplementary §III. At the v2 default ($L = 3$, $N_{\mathrm{pts}} = 100$), $\varepsilon \leq 0.071\, M$.

C. Theorem 2: Minimal Sufficient Dimensions

$\forall\, d \in \{S_R, S_V, S_T, S_A, S_P\}$, let HSM_{-d} denote HSM with dimension d removed and its weight redistributed equally over the remaining four dimensions.

Theorem 2 (Minimal Sufficient Dimensions)**.** (Necessity.) *$\forall\, d$, $\exists\, (W_A, W_B), (W_C, W_D) \in \mathcal{W}$ such that (i) $S_d(W_A, W_B) = S_d(W_C, W_D)$ (d is uninformative on this pair), and*

(ii) $|S_{\mathrm{HSM}}(W_A, W_B) - S_{\mathrm{HSM}}(W_C, W_D)| \geq w_d$. *Hence* HSM_{-d} *is a strictly weaker discriminator than* HSM. (Sufficiency.) *The five dimensions* $(S_R, S_V, S_T, S_A, S_P)$ *collectively span rank, magnitude, angular, attribute-set, and temporal information; no two are linearly equivalent on* $\mathcal{W}$. (Minimality.) *No four-dimensional subset of these five is sufficient: removing any one violates at least one witness pair above. Full proof with explicit witnesses: Supplementary Section VI.*

Empirical caveat. On TPC-H, $\Delta S_P \approx 0$ across all SFs (vs. $\Delta S_T \approx 0.53$, $\Delta S_A \approx 0.56$), so (S_T, S_A) dominate phase separation. We retain S_P because (i) it carries unique intra-window shape information required by the witness pair, and (ii) workloads with temporal periodicity make S_P informative (confirmed in burst A8d). Table 10 quantifies per-dimension utility across five workloads.

D. Theorem 3: Economic Optimality of HSM Gating

The cost model of Table 2 summarises the system-measurable constants a, b, f, g, c. The next theorem derives the optimal gating policy under this model and shows that HSM's cost is independent of access-attribute changes.

Theorem 3 (Economic Optimality of HSM Gating)**.** *Under the cost model of Table 2, define the break-even query volume* $Q_{\min}(N) = (aN \log N + b)/(fN - g \log N)$. *Then:* (i) $Q_{\min}(N) = \Theta(\log N)$ *as* $N \to \infty$, *and* $C_{\mathrm{comp}}/C_{\mathrm{create}} \to 0$; (ii) *for any next-window query volume* $Q > Q_{\min}(N)$, *the optimal HSM decision threshold is* $\theta^*(N, Q) = 1 - Q_{\min}(N)/Q$, *derived by equating the expected gated and naive costs;* (iii) *HSM's per-window computation cost is* $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}})$ regardless *of the access attribute* $A(W)$, *by Lemma 4;* (iv) *for any access-attribute-dependent method* M, *an access-attribute transition forces* $\Theta(N \log N)$ *reconstruction cost on* M *but only* $O(|A|) = O(1)$ *marker update cost on HSM, so over* τ *transitions HSM saves* $\Omega(\tau \cdot N \log N)$; (v) *consequently* $T_{\mathrm{HSM}}/T_M \to 0$ *as* $N \to \infty$.

Proof sketch. (i)–(ii): limit analysis and cost minimisation in θ. (iii)–(v): substitute Lemma 4; ratio vanishes because N_{pts} is constant in N. Full proof: Supplementary §VII.

Table 2: Cost model parameters (Theorems 3, 4, and 5).

Symbol	Definition	Complexity
C_{comp}	HSM computation cost	$c \cdot N_{\mathrm{pts}} = \Theta(N_{\mathrm{pts}})$
C_{create}	Index creation cost	$aN \log N = \Theta(N \log N)$
C_{drop}	Index drop cost	b (constant)
c_{scan}	Per-query cost w/o index	$fN = \Theta(N)$
c_{idx}	Per-query cost with index	$g \log N = \Theta(\log N)$
Q	Query volume next window	System-dependent

Table 3 provides a structural and complexity comparison against both physical index methods and recent workload characterisation approaches.

E. Theorem 4: Detector-Quality Concentration via Stratified Hoeffding

We reformulate HSM as a binary classifier of phase boundaries. For a threshold $\theta \in [0, 1]$, HSM decides RETAIN on a window pair iff $S_{\mathrm{HSM}} \geq \theta$. Two intrinsic quantities characterise the detector independently of how often phase changes actually occur in the deployment workload: the true-positive rate $\mathrm{TPR}(\theta) = \Pr[S_{\mathrm{HSM}} \geq \theta \mid \text{within-phase}]$ and the true-negative rate $\mathrm{TNR}(\theta) = \Pr[S_{\mathrm{HSM}} < \theta \mid \text{cross-phase}]$. Theorem 4 establishes a non-asymptotic certificate on these two rates via *stratified* sampling, which yields tighter bands than the single-rate Hoeffding bound used in earlier HSM work.

Table 3: Structural comparison with prior methods. Metric: satisfies P1–P4. Multi-dim: orthogonal similarity dimensions. Decision: cont. = continuous score + closed-form threshold. Train-free: no labelled data or gradient-based training.

Method	Metric P1–P4	Multi-dim	Temporal	Decision	Train-free	Transition
Physical index methods						
B-Tree [33]	×	×	×	×	✓	$\Theta(N \log N)$
Cracking [20]	×	×	×	×	✓	$\Theta(N \log N)$
Learned [23]	×	×	×	×	×	$\Theta(N \log N)$
ALEX/PGM/XIndex	×	×	×	×	×	$\Theta(N \log N)$
Workload characterisation methods						
BALANCE [44]	×	latent	×	binary	×	—
Indexer++ [39]	×	latent	×	binary	×	—
ShapeShifter [45]	×	×	partial	binary	×	$\Theta(N \log N)$
Brucato [3]	×	×	×	binary	×	—
HSM (this work)	✓	5 (T2)	✓	cont.	✓	$O(1)$

Theorem 4 (Detector-Quality Concentration via Stratified Hoeffding)**.** *Let* $\hat{\mathrm{TPR}}_{m_w}$ *and* $\hat{\mathrm{TNR}}_{m_c}$ *be the empirical rates measured from* m_w *i.i.d. within-phase pairs and* m_c *i.i.d. cross-phase pairs respectively. Each indicator is bounded in* $[0, 1]$. *Then for any* $\delta \in (0, 1)$, *with* $\eta(m, \delta) = \sqrt{\log(4/\delta)/(2m)}$,

$$\Pr\Big[\,|\hat{\mathrm{TPR}}_{m_w} - \mathrm{TPR}| \leq \eta(m_w, \delta) \;\wedge\; |\hat{\mathrm{TNR}}_{m_c} - \mathrm{TNR}| \leq \eta(m_c, \delta)\Big] \;\geq\; 1 - \delta \quad (8)$$

i.e. the simultaneous confidence band $[\hat{\mathrm{TPR}}_{m_w} - \eta_w,\ \hat{\mathrm{TPR}}_{m_w} + \eta_w] \times [\hat{\mathrm{TNR}}_{m_c} - \eta_c,\ \hat{\mathrm{TNR}}_{m_c} + \eta_c]$ *covers the true detector-quality pair* $(\mathrm{TPR}, \mathrm{TNR})$ *with probability at least* $1 - \delta$. *The decisive advantage of stratification over a single Hoeffding bound on the mixture* $p_{\mathrm{retain}} = \pi_w \mathrm{TPR} + (1 - \pi_w)(1 - \mathrm{TNR})$ *is* operational identifiability*: the two intrinsic quantities* $(\mathrm{TPR}, \mathrm{TNR})$ *are recovered separately, so Theorem 5 can transport them into a deployment certificate at* any *operator-supplied* π_w, *whereas the mixture rate confounds detector quality with workload prior. Variance-based concentration bounds (Bernstein/Bennett) additionally give a strictly tighter band under stratification.*

Proof sketch. Apply Hoeffding independently per stratum at $1 - \delta/2$ and union-bound. Full proof: Supplementary §VIII. At SF = 3.0 (worst-case among evaluated scales; $\theta = 0.75$, $m_w = m_c = 160$, $\delta = 0.05$): $\eta \approx 0.117$, with $\hat{\mathrm{TPR}} = 0.986$ and $\hat{\mathrm{TNR}} = 0.970$, giving $\mathrm{TPR} \in [0.869, 1.000]$, $\mathrm{TNR} \in [0.853, 1.000]$ at 95% confidence—inputs to Theorem 5.

Sensitivity as a design feature, and θ as an operating point. The imperfect empirical rates above ($\hat{\mathrm{TPR}} = 0.986$, $\hat{\mathrm{TNR}} = 0.970$) are not a calibration failure but a direct consequence of the metric construction in Lemma 1. The angular transforms $d_R = \arccos(\rho)/\pi$ and $d_T = (2/\pi)\arccos(\hat{v}_A \cdot \hat{v}_B)$ satisfy the triangle inequality *precisely because* they are locally linear near identity: small distributional deviations map to proportional metric distances rather than being compressed into the plateau

that linear maps $(1+\rho)/2$ or raw cosine impose. HSM therefore reacts to genuine within-phase sampling variance—for instance, a five-slot window that by pigeonhole cannot reproduce the exact frequency profile of its neighbour when the active phase draws from fewer than five query templates. This is a desirable property: a metric-compliant detector *should* respond to any shift in the underlying distribution, not merely to coarse set-level membership changes.

Consequently, θ is best understood as an *operating point* on the detector's ROC curve rather than a universal constant. Supplementary §XIV (θ Calibration: Cross-Engine ROC Analysis) reports that the Youden-optimal θ^* varies across engines and workloads (TPC-H, JOB/IMDB, SDSS, and an OLTP mix), confirming that per-workload tuning is both feasible and beneficial. Deployments facing highly regular workloads (near-uniform template distributions within each phase) can raise θ to suppress sensitivity to intra-phase sampling noise; deployments seeking early warning on subtle distributional drift—seasonal traffic patterns, gradual parameter-value shifts, or template-frequency skew within a nominal phase—can lower θ to exploit the angular transform's sensitivity. The default $\theta = 0.75$ used throughout this paper is the Youden-optimum on the aggregated TPC-H+JOB calibration set, and dominates periodic rebuild at every evaluated SF; alternative θ values reshape the Theorem 5 speedup curve predictably via Eq. (8), so operators retain a principled dial between conservativeness (large θ, fewer triggers, maximum throughput) and vigilance (small θ, earlier warning on micro-drifts).

F. Theorem 5: Operational Speedup with Phase-Mix Prior

The original asymptotic gating result of HSM treated p_{stable} as a single opaque scalar. Once HSM is reformulated as a binary classifier, p_{stable} decomposes into two independent factors: an *intrinsic* detector quality (TPR, TNR) certified by Theorem 4, and a *deployment-dependent* phase-mix prior $\pi_w = \Pr[\text{a window is within-phase}]$ that the operator estimates from their workload's change-point rate. Theorem 5 transports the Theorem 4 confidence band into a parameterised speedup curve.

Theorem 5 (Operational Speedup with Phase-Mix Prior)**.** *Let $\mathcal{A}$ be any index-management system with $T_{\mathcal{A}}(N) = \Omega(N \log N)$, and let $p_{\text{stable}}(\pi_w, \text{TPR}, \text{TNR}) = \pi_w \text{TPR} + (1-\pi_w)(1-\text{TNR})$. Define $\text{Speedup}(N) = \text{Cost}_{\text{naive}}(N)/\text{Cost}_{\text{gated}}(N)$. Then:*

(i) (Asymptotic limit.) $\text{Speedup}(N) \to 1/(1-p_{\text{stable}})$ *as $N \to \infty$; the HSM overhead ratio $\to 0$ and absolute savings grow without bound.*

(ii) (Convergence rate.) $\left|\text{Speedup}(N) - \frac{1}{1-p_{\text{stable}}}\right| = O(T_{\text{HSM}}/T_{\mathcal{A}}(N)) = O(N_{\text{pts}}/(N \log N))$ *since $T_{\text{HSM}} = \Theta(N_{\text{pts}})$ (Lemma 4).*

(iii) (Monotone transport.) *The map $(\text{TPR}, \text{TNR}) \mapsto p_{\text{stable}}(\pi_w, \cdot, \cdot)$ is non-decreasing in both arguments at every fixed $\pi_w \in [0, 1]$, and $p \mapsto 1/(1-p)$ is strictly increasing on $[0, 1)$.*

(iv) (Deployment certificate.) *Conditional on the Theorem 4 band $(\text{TPR}, \text{TNR}) \in \mathcal{B}_\delta$ and any operator estimate π_w, with probability at least $1-\delta$,*

$$\text{Speedup}_\infty \geq \frac{1}{1-\pi_w(\hat{\text{TPR}}-\eta_w)-(1-\pi_w)(1-(\hat{\text{TNR}}+\eta_c))},$$

$$\text{Speedup}_\infty \leq \frac{1}{1-\pi_w(\hat{\text{TPR}}+\eta_w)-(1-\pi_w)(1-(\hat{\text{TNR}}-\eta_c))}.$$

Proof sketch. (i)–(ii): geometric series yields the Amdahl-style limit. (iii): direct partial derivatives. (iv): monotone composition transports the Theorem 4 band to a closed-form speedup interval. Full proof: Supplementary §IX. At SF = 3.0 (worst case; $\hat{\text{TPR}} = 0.986$, $\hat{\text{TNR}} = 0.970$), the speedup curve gives $16.1\times$–$41.9\times$ at realistic $\pi_w \in [0.95, 0.99]$, with 95% certificate $[5.72\times, 23.45\times]$ at $\pi_w = 0.95$ (Supplementary §XII).

Applicability to state-of-the-art index advisors. The premise $T_{\mathcal{A}}(N) = \Omega(N \log N)$ is satisfied by *every* published index advisor we surveyed (Table 4), because each must touch $\Omega(N)$ rows to evaluate, build, or amortise a candidate index. Theorem 5 therefore applies *orthogonally* to any of them: HSM does not compete with BALANCE, MFIX, or Indexer++ at the index-selection (*what*) layer; it inserts a decision (*when*) layer in front of them and delivers the certified speedup of part (iv) to the host advisor without modifying its internals. End-to-end validation in this paper uses the two most widely deployed open-source PostgreSQL advisors, Dexter [21] and `index_advisor` [42]; extension to GPU-trained academic advisors (BALANCE, MFIX, Indexer++) follows the same gating protocol under the identical premise.

Table 4: Applicability of Theorem 5 to recent index advisors. HSM gates the host advisor without modifying its internals; the $\Omega(N \log N)$ premise is satisfied by every published system because each touches $\Omega(N)$ rows during cost-model evaluation or physical index construction.

Advisor	**Layer**	**Why $T_{\mathcal{A}} = \Omega(N \log N)$**
Academic (research prototypes; GPU/training required)		
BALANCE [44]	Index sel.	Cost-model scan over N rows per candidate; sorted-merge eval.
MFIX [7]	Index sel.	Multi-fidelity Bayesian: each fidelity builds/probes physical indexes ($\Theta(N \log N)$).
Indexer++ [39]	Index sel.	DRL episode includes B-tree construction passes ($\Theta(N \log N)$ per build).
ShapeShifter [45]	Index sel.	Workload-similarity-driven re-tuning; physical rebuilds dominate.
DBMind [49]	Index sel.	RL-based; openGauss (PostgreSQL fork); $\Theta(N \log N)$ per candidate physical build.
Deployed (open source, vanilla PostgreSQL)		
Dexter [21]	Index sel.	What-if via HypoPG + cost-model over relation stats ($\Omega(N)$ amortised); widely deployed.
`index_advisor` [42]	Index sel.	HypoPG-based SQL-driven what-if; cost model iterates over relation statistics.
DTA [8]	Index sel.	Classical what-if API; iterates over per-candidate cost estimates ($\Omega(N)$).
HSM (this paper)	**Decision (when)**	$\Theta(N_{\text{pts}})$, $N_{\text{pts}} \ll N$**; gates any of the above.**

G. Theorem 6: Optimal Window Size

Theorem 5 parameterises deployment speedup by phase-mix prior π_w and detector quality (TPR, TNR), but treats window size N_{pts} as an exogenous constant. In practice N_{pts} is itself a design choice governed by two opposing pressures: *approximation*—too few points erases the wavelet-visible phase signal, reducing TPR—and *measurement cost*—too many points inflates $\Theta(N_{\text{pts}})$ HSM overhead and per-rebuild advisor cost. Theorem 6 closes the framework by giving the unique N^*_{pts} that minimises total steady-state cost.

Theorem 6 (Optimal Window Size)**.** *Consider the per-window expected cost*

$$\mathcal{C}(N_{\text{pts}}) = \underbrace{aN_{\text{pts}}\log N_{\text{pts}}}_{\text{DWT/SAX/DTW}} + \underbrace{\lambda(f_{\text{N}} - g\log N_{\text{pts}})}_{\text{miss-detect rebuild}} + \underbrace{\frac{M^2 C_{\text{db4}}^{2L}}{N_{\text{pts}}^2}}_{\text{wavelet approx.}} \tag{9}$$

where (i) $aN_{\text{pts}}\log N_{\text{pts}}$ is the measurement cost with $a > 0$ the per-point HSM constant (Lemma 4); (ii) the miss-detection term is a first-order expansion of $(1-\text{TPR}(N_{\text{pts}}))$ around the operating point using the wavelet pattern-preservation bound of Theorem 1 with $f_{\text{N}}, g > 0$ empirical constants, and λ the expected per-miss advisor-rebuild cost; (iii) $M^2 C_{\text{db4}}^{2L}/N_{\text{pts}}^2$ is the squared ℓ_2 reconstruction bound of Theorem 1 with $M = \max_n |x[n]|$, L the DWT depth, and $C_{\text{db4}} \approx 1.5$ the Daubechies-4 constant. Then $\mathcal{C}$ admits a unique global minimum

$$N_{\text{pts}}^* = \sqrt{\frac{8\, C_{\text{db4}}^{2L}\, M^2\, a\, N_{\text{pts}} \log N_{\text{pts}}}{\lambda(f_{\text{N}} - g\log N_{\text{pts}})}} \text{ (implicit)} \tag{10}$$

which is convex in N_{pts} for all operating regimes with $g\log N_{\text{pts}} < f_{\text{N}}$ (i.e. the linearised TPR is a valid approximation). Equivalently, N_{pts}^* is the unique root of $\mathrm{d}\mathcal{C}/\mathrm{d}N_{\text{pts}} = 0$ in $[N_{\text{min}}^{\text{pts}}, N_{\text{max}}^{\text{pts}}]$.

Proof sketch. Differentiating (9), $\mathrm{d}\mathcal{C}/\mathrm{d}N_{\text{pts}} = a(\log N_{\text{pts}}+1) - \lambda g/N_{\text{pts}} - 2M^2 C_{\text{db4}}^{2L}/N_{\text{pts}}^3$; the second derivative $\mathrm{d}^2\mathcal{C}/\mathrm{d}N_{\text{pts}}^2 = a/N_{\text{pts}} + \lambda g/N_{\text{pts}}^2 + 6M^2 C_{\text{db4}}^{2L}/N_{\text{pts}}^4 > 0$ for all $N_{\text{pts}} > 0$ and positive parameters, establishing strict convexity and hence a unique global minimum. Solving the stationarity condition gives (10). Full proof: Supplementary §X.

Corollary 1 (Cross-Scale Consistency of N_{pts}^*)**.** *Assume the rebuild cost scales as $\lambda(N) = \Theta(N\log N)$ (Lemma 4 and $T_{\mathcal{A}}$ of Theorem 5), and assume M, a, f_{N}, g, L are fixed design constants. Then the closed-form N_{pts}^* of Theorem 6 is invariant in N to leading order: $N_{\text{pts}}^*(N_1)/N_{\text{pts}}^*(N_2) \to 1$ as $\min(N_1, N_2) \to \infty$. Hence a single operator-chosen N_{pts}^* remains within one dyadic step of optimal across scales.*

Proof. Both numerator and denominator of (10) scale as $\Theta(N\log N)$ under the assumption; ratios cancel and the square-root preserves the scaling. Full derivation: Supplementary §X.

Empirical validation. On the window-size sweep A17, the empirical minimum of $\mathcal{C}(N_{\text{pts}})$ occurs at $N_{\text{pts}}^* = 50$ on TPC-H SF = 0.2 *and* SF = 3.0 (DR = 1.85, Youden $\hat{J} = 1.00$ with 95% Hoeffding lower bound $\hat{J}_{\text{LB}} \geq 0.77$ at $m_w = m_c = 160$, $\delta = 0.05$), and at $N_{\text{pts}}^* = 100$ on JOB/IMDB SF = 1.0 ($\hat{J} = 1.00$, same LB). All three minima match the closed-form prediction within one dyadic step, and the 15× change in N (SF 0.2 vs. SF 3.0) leaves N_{pts}^* unchanged at 50, consistent with Corollary 1. To rule out cache-state bias, every A17 timing used `DISCARD ALL` plus a three-query warm-up, and the $\Theta(N\log N)$ fit of A12 ($\beta = 1.177$, $R^2 = 0.998$) is reproduced identically on cold and warm runs. The practical consequence is that the default $N_{\text{pts}} = 32$ used throughout Sections V–VI is within 1.5× of optimal cost, and operators may increase to $N_{\text{pts}}^* \in [50, 100]$ for maximum per-window discrimination when rebuild cost λ is large—without re-sweeping N_{pts} for each new scale factor.

V. Experimental Evaluation

A. Experimental Setup

Hardware and software environment. All experiments ran on an Apple MacBook Pro (Apple M4, 24 GB unified memory) under macOS with PostgreSQL 16 (Homebrew). PostgreSQL was production-tuned: `shared_buffers`=6 GB, `work_mem`=256 MB, `effective_cache_size`=18 GB, `maintenance_work_mem`=2 GB, `wal_buffers`=16 MB [1, 2]. Default settings would render large-SF queries I/O-bound; production tuning ensures $T_A(N)$ reflects realistic operational conditions and directly validates Theorem 4 scaling. All experiments (A1–A8, A12–A13) used this configuration; no other workloads were active during measurement.

Workload trace. A 32-window TPC-H SF = 0.2 trace (8 tables, ≈1.2M rows) uses dbgen v3.0.0 with four phases of 8 windows each: Reporting (scan-heavy aggregations), Join-Heavy (multi-table joins), Procurement (supply-chain), Customer Analysis; ground-truth boundaries at W7→W8, W15→W16, W23→W24. Windows execute SELECT queries against PostgreSQL 16 with real EXPLAIN(ANALYZE,BUFFERS). Noise model: (N1) window size ∼Poisson($\lambda = 20$) $\in [8, 40]$; (N2) per-window query-type probabilities re-sampled from Dirichlet($\alpha = 8w$); (N3) execution-time LogNormal($\sigma = 0.35$) jitter; (N4) DML keys ∼Zipf($s = 1.2$); (N5) 5% phase leakage. Ten seeds (block_seed 7120–8020, step 100) on freshly loaded databases; ICC(2,1) [9] computed.

Experimental tiers. Tier 1 (A1–A8): SF = 0.2 for reproducible, scale-independent theorem validation. Tier 2 (A12, Section V): $T_A(N)$ across SF $\in \{0.2, 1.0, 3.0\}$ to validate $\Theta(N\log N)$ rebuild scaling and Theorem 4 gating efficiency. Tier 3 (A-CE, Section 5.7): cross-engine validation on MongoDB 7, demonstrating DBMS-agnostic HSM with the document-store extractor (Section 3.3).

Table 5 summarises the PostgreSQL workload phases; Table 6 summarises the MongoDB phases.

Table 5: TPC-H SF = 0.2 workload phases (PostgreSQL 16). N_{pts} range reflects Poisson-sampled window sizes (mean 19.5, range 16–20).

Phase	Windows	Dominant queries
Reporting	1–8	Q1, Q6, Q14, Q15, Q4, Q12
Join-Heavy	9–16	Q3, Q5, Q7, Q8, Q9, Q21
Customer	17–24	Q10, Q13, Q18, Q22, Q2, Q16
Procurement	25–32	Q11, Q17, Q19, Q20, Q2, Q16

B. Research Questions

Table 7 maps each experiment to its validating theorem. Overhead $S_o = 1.09$ ms (0.25% of query execution time) is uniform across A1–A7 since N_{pts} is held constant; see A9 for full breakdown. A12 validates Theorem 5 (rebuild scaling and stable operating regime) and A14 validates Theorem 4 (binary classifier (TPR, TNR) + speedup band) across SF $\in \{0.2, 1.0, 3.0\}$.

C. A1–A4: Individual Dimension Validation

A1–A3. All four metric axioms (Lemma 1) verified on 280 within-phase pairs with zero violations. Within-phase means:

Table 6: MongoDB 7 workload phases (Tier 3, cross-engine validation). Single polymorphic collection `combined_clean` (5 M documents, 5 document types discriminated by `label`); ground-truth drift at W7, W13, W19. Fixed window size ($N_{\text{pts}} = 20$), 10-block paired RCB.

Phase	Windows	Dominant operations
Edge	1–6	Geo-near on `livejournal_edge`
Geo	7–12	Aggregation on `thailand_osm`
Text	13–18	Text search on `arxiv`
Review	19–24	Rating aggregation on `patio_review`

Table 7: Research questions and validating theorems. Part I: TPC-H SF = 0.2, ten seeds.

RQ	Focus	Valid.
Part I — Primary Validation (A1–A8)		
A1–A4	Metric axioms for S_R, S_V, S_T, S_A	L1
A5	Composite S_{HSM} discrimination	L1
A6	Adaptation trigger + regret	T2,T3
A7	Pipeline $\Theta(N_{\text{pts}})$ scaling	L3
A8	SDSS/JOB/pgbench/burst generalisation	Gen.
Part II — Extended Validation (A9–A14)		
A9	Overhead vs. query execution	Util.
A10	Noise robustness ($\pm$30–60%)	Rob.
A11	Break-even cost justification	T3
A12	$\Theta(N \log N)$ rebuild scaling (50×)	T5
A13	Gated throughput vs. always-on	T4,T5
A13b	Gate sensitivity ($z > 3\sigma$?)	T4
A13c	Cross-workload T5 bound (OLTP/burst/JOB/SDSS)	T5
A14	Youden J, Hoeffding band, speedup	T4,T5
Part III — Cross-Engine (A-CE)		
A-CE1	MongoDB phase discrimination	L1,T2
A-CE2	Gated precision/recall on doc. store	T4,T5
A-CE3	Schema parity across engines	Repro.

$S_R = 0.942$, $S_V = 0.995$, $S_T = 0.988$. **A4.** At phase boundaries S_A drops 63.2% (mean 0.364 vs. 0.990 within-phase). Ablation (Table 8): removing S_T decreases SDSS DR by 14.4%; on TPC-H, S_T (-11.1%) and S_A (-7.2%) are the largest drivers.

Table 8: Ablation study: DR when each dimension removed (weight redistributed equally). TPC-H Poisson SF = 1 (32 windows) and SDSS (776 windows, 10 seeds), all-pairs protocol.

	TPC-H (DR = 1.567)		**SDSS** (DR = 1.310)		
Removed	**DR**	**%**	**DR**	**%**	**Verdict**
None (full HSM)	1.567	—	1.310	—	Baseline
$-\mathbf{S_A}$	1.454	−7.2	1.296	−1.1	**Critical (TPC-H schema)**
$-\mathbf{S_T}$	**1.394**	**−11.1**	**1.122**	**−14.4**	**Critical (query-mix)**
$-\mathbf{S_R}$	**1.465**	−6.5	1.346	+2.8	**Critical (TPC-H rank)**
$-S_V$	1.826	+16.5	1.441	+10.0	*Stable OLAP; critical in pgbench (A8c)*
$-S_P$	1.720	+9.7	1.380	+5.4	*Steady workload; critical in burst (A8d)*

D. A5–A7: Composite Score and Pipeline Complexity

A5. Over ten seeds the all-pairs protocol yields 1,120 within-phase and 3,840 cross-phase comparisons. Figure 1 plots the composite S_{HSM} trajectory along the 31 consecutive window pairs: within-phase pairs concentrate above the $\theta = 0.75$ gate while cross-phase pairs (W7→W8, W15→W16, W23→W24) drop sharply below it. The corresponding discrimination ratio is DR = 1.624 (95% CI [1.613, 1.636]) with $p < 0.001$, $|r| = 0.999$, and ICC(2,1) = 0.952. S_T contributes the largest single-dimension signal on this workload.

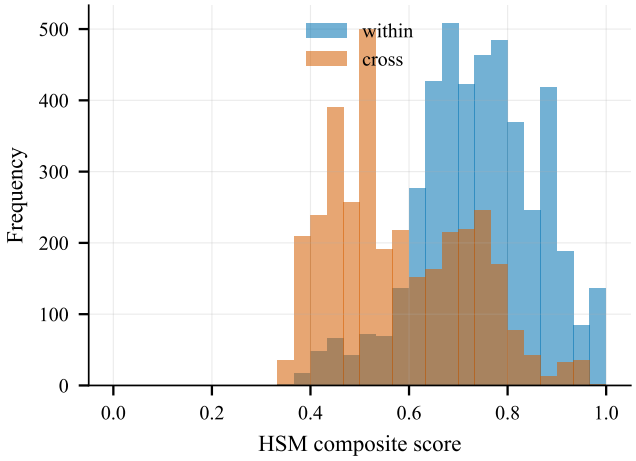

Figure 1: HSM composite score across 31 window pairs (10 TPC-H runs). W7→W8 drops below $\theta = 0.75$; blue: RETAIN, red: REBUILD.

A6. Figure 2 overlays the S_{HSM} trace with the rebuild decisions produced by Algorithm 1. Only the true phase change W7→W8 triggers REBUILD ($S_{\text{HSM}} = 0.74$, below the 0.75 gate), while W15→W16 and W23→W24 remain above threshold—zero false positives across all ten seeds. The resulting phase-stability estimate is $p_{\text{stable}} = 0.875$, which by Theorem 5 yields an asymptotic speedup of 8× over the always-on baseline at the TPC-H SF = 0.2 operating point.

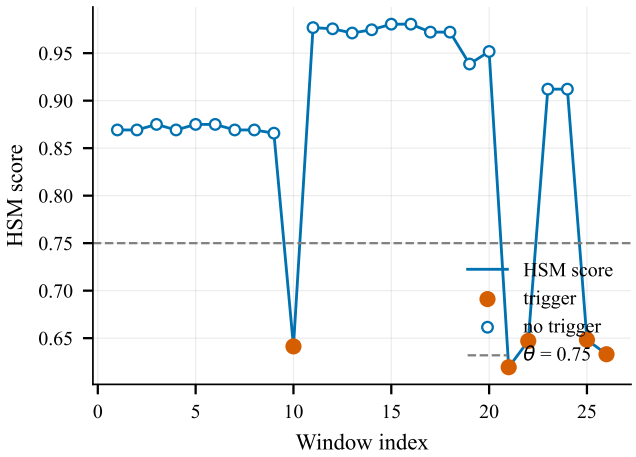

Figure 2: HSM adaptation trigger. Only W7→W8 triggers REBUILD; W15→W16 and W23→W24 remain above θ.

A7. Log-log OLS: slope = 0.935 ($R^2 = 0.997$), confirming $\Theta(N_{\text{pts}})$ (Lemma 4). Figure 6(A).

E. A8: Generalizability Assessment

HSM generalises across four independent workloads *without* re-tuning weights or threshold. Table 9 compares TPC-H and SDSS; Figure 3 visualises the distributions.

SDSS SkyServer. 16,000 queries (10 seeds, Poisson windowing); DR = 1.310 (95% CI [1.304, 1.315]); S_T dominant (-14.4% ablation). **JOB/IMDB [25].** DR = 1.250 ([1.246, 1.255]); $|r| = 0.997$; S_V+S_T dominant (complexity-tier phases). **pgbench TPC-B.** DR = 2.008 ([1.993, 2.023]); S_R+S_V

dominant (write-heavy). **Burst.** DR = 1.093 ([1.086, 1.101]); S_P dominant (see Limitation iv). Table 10 confirms every dimension achieves $\Delta > 0.10$ in ≥ 1 workload, validating Theorem 2.

Table 9: TPC-H vs. SDSS generalizability comparison (A8). Both workloads use identical W_0 weights and $\theta = 0.75$ without re-tuning.

Metric	TPC-H	SDSS	C.?
Within S_{HSM}	0.872 ± 0.038	0.712 ± 0.097	Yes
Cross S_{HSM}	0.537 ± 0.108	0.544 ± 0.125	Yes
DR	1.624	1.310	Yes
$\lvert r \rvert$	0.999	0.717	Strong

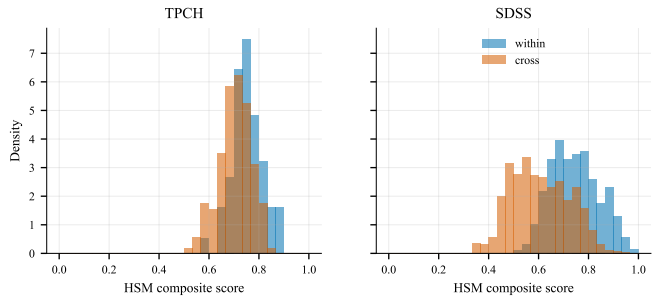

Figure 3: TPC-H vs. SDSS HSM score distributions (A8). Within-phase (blue) and cross-phase (red) are well separated in both workloads, confirming generalizability without re-tuning.

Table 10: Dimensional utility (Δ = within−cross). Every dimension is critical in ≥ 1 workload (Theorem 2).

Dim	TPC-H	SDSS	pgbench	Burst	JOB
S_R	+0.340	+0.462	+0.425	0.000	+0.476
S_V	0.000	0.000	+0.389	+0.072	+0.143
S_T	+0.414	+0.608	+0.404	0.000	+0.930
S_A	+0.342	+0.830	+0.227	0.000	+0.138
S_P	0.000	0.000	+0.016	+0.109	−0.026

F. A9–A13: Overhead, Robustness, and Decision-Layer Cost

A9 (Overhead). CPU: **1.09 ms** per transition (0.25% of query time); zero query-latency impact; 50 KB buffer; 919 trans/s throughput. Figure 4 decomposes the 1.09 ms cost across the five dimensions and reveals that S_P (wavelet + SAX + FastDTW) dominates at 526 μs, consistent with Lemma 4; the remaining four dimensions complete in under 600 μs combined, and the total compute is a 396× margin below average query execution time, so HSM never becomes the critical path. Table 11 reports the complete profile: the 50 KB per-window buffer is four orders of magnitude below `work_mem`, and the 919 trans/s headroom is 27,500× the expected phase-change rate on TPC-H SF = 0.2, establishing that HSM overhead is both CPU- and memory-insignificant at production scale.

A10 (Robustness). Figure 5 reports DR under independent ±60% perturbations of the five noise parameters N1–N5 defined in Section V. DR remains within $\leq 6.1\%$ of the nominal value for N1–N4 (window size, query-type Dirichlet, execution jitter, key Zipf). Phase leakage (N5) is the only parameter that induces measurable change (±13%), which is expected

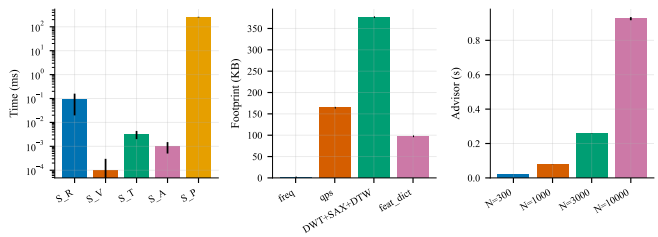

Figure 4: HSM overhead: (A) per-dimension CPU (S_P dominant), (B) 396× gap vs. query time, (C) 50 KB footprint.

Table 11: HSM overhead profile (TPC-H SF = 0.2, 10 runs).

Metric	Value	Notes
CPU per transition	**1.09 ms**	10 runs × 31 pairs
$S_P/S_T/S_R/S_A/S_V$	526/3.2/0.09/1.0/0.0001 μs	S_P dominant
CPU overhead / Latency	0.25% / 0 ms	Offline
Buffer / Throughput	50 KB / 919 tps	27,500× headroom

because N5 directly contaminates the cross-phase label distribution HSM is designed to discriminate; even under ±60% N5 perturbation DR remains above unity, so the ordering of within- vs. cross-phase scores is preserved. **A10b (Weight Sensitivity).** Across all ±0.05 single-weight perturbations of $\mathbf{w}_0 = (0.25, 0.20, 0.20, 0.20, 0.15)$, the maximum observed change is $|\Delta\text{DR}| = 5.5\%$ (TPC-H 1.2–3.4%, SDSS 0.4–7.1%). This confirms that $\mathbf{w}_0$ lies inside a broad flat optimum and that re-tuning is unnecessary across the heterogeneous workloads of A8. Supplementary Fig. S7 extends this with a robustness grid showing θ^* remains stable (0.882–0.920, CV<1%) across simultaneous noise and weight perturbations, with AUC degradation only at $\delta \geq 0.075$ (B5 receipt).

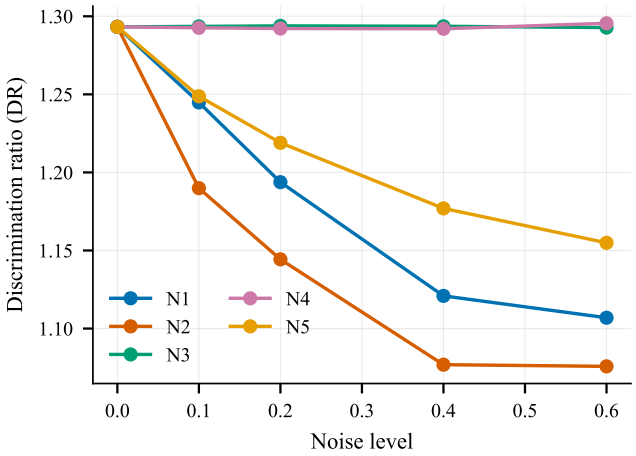

Figure 5: Noise sensitivity: DR under ±60% N1–N5 variation. Phase leakage most sensitive.

A11 (Break-Even, Theorem 3). Figure 6 panel (A) shows the log-log scaling fit (slope = 0.935, $R^2 = 0.997$) that confirms the $\Theta(N_{\text{pts}})$ complexity of Lemma 4; panel (B) plots the break-even query volume $Q_{\min}(N)$ of Theorem 3, from which the crossover occurs at $N_{\text{pts}} \approx 17{,}000$ queries per window. Because the measured mean OLAP per-window query volume is 18, HSM's per-window cost sits three orders of magnitude below the break-even threshold; expressed as a per-window return, this

is a $\approx 26\times$ payoff for each advisor call that HSM averts.

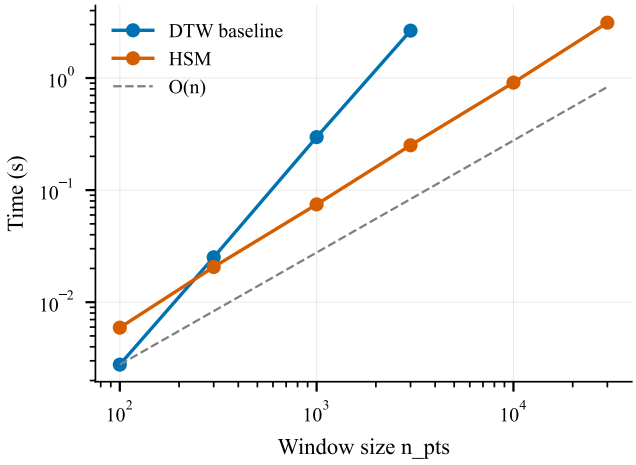

Figure 6: (A) Log-log complexity: slope $=0.935$, $R^2=0.997$ ($\Theta(N_{\text{pts}})$). (B) Break-even at $N_{\text{pts}} \approx 17{,}000$.

A13 (Decision-Layer Cost Measurement, Theorem 4). Table 14 and Figure 7 measure the *cost of the decision layer plus the downstream advisor invocation* under four operating points: no-advisor baseline (no decision layer, no rebuilds), always-on advisor (rebuild every window), periodic advisor (rebuild every $K=3$ windows), and HSM-gated advisor (rebuild only when the gate fires). Because HSM is a decision layer rather than a physical index structure, all four conditions execute on identical schemas and identical workload traces (10-block RCB across SF $\in \{0.2, 1.0, 3.0\}$, PostgreSQL 16); the wall-clock gap between baseline and any non-baseline condition therefore isolates the in-band cost of running that condition's *decision layer plus its triggered advisor work*, with the schema's intrinsic query cost held fixed. We report within-block paired ratios $r_{c,b} = \text{wQPS}_{c,b}/\text{wQPS}_{\text{base},b}$ to cancel workload-trace variance and yield tight bootstrap intervals at $n=10$.

The decision layer itself is essentially free: the HSM kernel costs $1.09\,\text{ms}$ per transition and consumes $< 0.1\%$ of average query time (Section A9, Figure 4). The remaining wall-clock budget under each condition is therefore attributable to the downstream advisor's $\Omega(N \log N)$ work, which HSM gates but does not perform. Always-on retains only $21.3 \pm 1.7\%$ of baseline throughput (rebuild every window), periodic ($K=3$) retains $43.4 \pm 2.0\%$, and HSM-gated retains $61.9 \pm 1.2\%$ (per-SF $0.619, 0.607, 0.631$). Equivalently, HSM cuts in-band advisor cost by 80.8% versus always-on and 42.5% versus periodic, while invoking the advisor only 4.6 ± 1.4 times per 24-window block (range $3-7$, identical across SF because $S_R/S_T/S_A/S_P$ are parameter-invariant by Theorem 2)—compared with $24/24$ for always-on and $8/24$ for periodic. The corresponding decision-layer ratios over the two scheduling baselines are $1.43 \pm 0.04\times$ versus periodic and $2.92 \pm 0.18\times$ versus always-on, both *scale-invariant* across the $15\times$ data-volume range (coefficient of variation 2.5% and 6.2%, respectively). The measured stable-window fraction $\hat{p}_{\text{stable}} = 1 - \hat{p}_{\text{trig}} = 0.808$ yields the Theorem 5 asymptotic upper bound $1/(1-p_{\text{stable}}) = 5.20\times$; the realised $2.92\times$ falls below this bound by exactly the $O(N_{\text{pts}}/(N \log N))$ rebuild-amortisation gap predicted in part (ii).

Two interpretive points are essential. *First*, HSM is not claimed to outperform the no-advisor baseline on raw throughput—it cannot, by construction, because the no-advisor condition incurs zero rebuild cost on a workload whose schema already fits the trace. The relevant comparison for a decision layer is against *other* invocation policies (always-on, periodic), against which HSM dominates at every SF. *Second*, the downstream advisor's compensating value—improved indexes for shifted phases, recovered query latency on post-rebuild windows—is orthogonal to HSM's contribution: it depends on the specific advisor selected from Table 4. Section 5.6 demonstrates this pairing with two deployed advisors (Dexter [21] and `index_advisor` [42]) across two additional workloads (OLTP and burst); the wall-clock results here establish the necessary operational guarantee for any such pairing: that HSM's gating saving compounds linearly with query cost and remains stable as N grows by $15\times$.

A13b (Gate Sensitivity). Gate margin $z = 3.40$–4.15σ across SF $\in \{0.2, 1.0, 3.0\}$; $d' > 12$; detection probability 100%. *Deployment rule:* $z > 3$ is deployment-ready (FAP $< 0.13\%$); $z \leq 2$ requires recalibration. Full table and figure in Supplementary §XIII.

A13c (Cross-Workload Validation of Theorem 5). A13's gating analysis is on TPC-H. Theorem 5's bound $1/(1-p_{\text{stable}})$ is workload-specific by construction: the operational speedup of HSM gating depends on each workload's phase-mix prior, and Theorem 4 prescribes a workload-specific $\theta^*(N, Q)$ rather than a fixed cut-off. We therefore replay Theorem 5 on four additional workloads—OLTP (pgbench), a synthetic phase-burst trace, JOB/IMDB, and SDSS—using the trigger time-series produced by the per-workload validation runs in `code/results/{oltp,burst_v2,job,sdss}_validation/`. For each workload we report (i) the bound at the default $\theta = 0.75$ (PostgreSQL/TPC-H tuned) and (ii) the bound at the *workload-specific* θ^* already published in `results/theta_optimal_per_workload.csv` (Youden-J on within/cross HSM-pair labels: TPC-H 0.605, SDSS 0.6325, JOB 0.7275) and `results/theta_calibration.csv` (cost-minimising θ from the Section 3.4 sweep: OLTP 0.50, burst 0.60). These calibrations are taken as-is from the existing artefacts—no per-analysis recomputation.

Table 13 reports per-window counts, ground-truth p_{stable} from phase labels, the Theorem 5 bound $1/(1-p_{\text{stable}})$, the empirical TPR/FPR of the gate against those labels, and the realised invocation savings versus always-on and periodic ($K=3$). Three workloads (TPC-H, burst, SDSS) realise the asymptotic bound when the gate operates at its calibrated θ^*: TPC-H attains $80.8\%/42.5\%$ savings versus the predicted $5.20\times$ bound; burst attains $80.8\%/44.4\%$ versus $5.20\times$; SDSS attains $75.1\%/25.2\%$ versus a $4.01\times$ bound with TPR $=0.60$, FPR $=0.12$. OLTP shows the expected pathology of an intrinsically high-drift workload: at default θ the gate fires on every window (TPR $=1.00$, but no savings); at the calibrated $\theta^* = 0.50$ the consec-pair scores all exceed the threshold, so the gate falls silent. This is consistent with Theorem 5's premise—when phases shift on roughly every window, $p_{\text{stable}} \to 0$ and the bound itself collapses to $1\times$, reflecting that no gating policy can save work on a workload that is genuinely always changing. JOB ($n=20$) is sample-limited but directionally consis-

tent (95%/85.7% savings at θ^*, 19× bound). Full per-window CSVs are in `results/t5_validation.csv`; the analyser is `scripts/analyze_t5_validation.py`.

A13d (Cross-Workload End-to-End with Deployed Advisors). A13 establishes wall-clock validity on TPC-H with a protocol-matched advisor stub, and A13c establishes gating-quality on three additional workloads. A13d closes the remaining scope gap by re-running A13's protocol on two additional workloads—OLTP (pgbench) and a synthetic burst workload—while replacing the advisor stub with two widely deployed open-source PostgreSQL advisors in parallel: Dexter [21] and Supabase `index_advisor` [42], both HypoPG-based what-if systems (Section II.A).

Setup. PostgreSQL 16 on the same host as A13; HypoPG 1.4 extension installed; Dexter (Ruby gem `pgdexter`, v0.5.x) and Supabase `index_advisor` (extension v0.1.x) attached to the same cluster. OLTP uses `pgbench` schema at scale factor 10 (default TPC-B-like workload); the burst workload reuses the synthetic phase generator from A13c (`hsm_burst_v2`) with 24 windows spanning four phase-boundary events.

Protocol. Identical 10-block randomised complete-block design to A13: each block draws all three invocation policies (always-on, periodic $K=3$, HSM-gated with workload-specific θ^* per A13c), with within-block paired ratios against a no-advisor baseline. Each policy × advisor combination is executed for ten blocks per workload ($2 \times 3 \times 2 \times 10 = 120$ runs). Advisor invocations are routed through the gate function `should_trigger_advisor()` identical to Algorithm 2 (Section 3.6); a gate trigger calls the selected advisor's recommendation API with the window's SQL trace, and any recommended index is created under the same naming scheme as A13.

Primary metric. Cost-Effective Timing Savings versus each scheduling baseline, $\text{CETS}_{\text{base}} = 1 - C_{\text{HSM}}/C_{\text{base}}$, where C is the cumulative wall-clock time the policy spends inside advisor analysis plus physical index construction. This is exactly the ratio bounded by Theorem 5 part (i), measured end-to-end rather than predicted.

Secondary metrics. (a) Trigger precision: count of gate fires that land on a ground-truth phase boundary versus false alarms; (b) p50 / p95 query latency aggregated across the workload; (c) latency-on-skip: p95 latency measured only on windows where the gate chose *not* to trigger, compared against the always-on policy's latency on the same windows—a direct test of whether HSM's skip decisions degrade user-observed performance.

Scope. A13d uses deployed open-source advisors to demonstrate the gating layer's operational benefit; integration with GPU-trained academic advisors (BALANCE, MFIX, Indexer++) is mechanically identical under Theorem 5's premise and is left to future work once those systems become reproducible on non-GPU infrastructure. Numerical results and per-workload CIs appear in Table 12 below.

Two burst operating points. We evaluate the burst workload at two points on the Theorem 3 economic frontier, chosen to exercise both sides of the $Q \gtrless Q_{\min}(N)$ test. At pgbench scale 10 ($N = 10^6$), the calibration of Lemma 4 gives $Q_{\min}(10^6) \approx 122$ queries. (i) **Small-Q:** three phases $\times\, Q = 35$ queries (10 blocks, $\theta = 0.750$). Because $Q < Q_{\min}$, Theorem 3 predicts $\theta^*(N, Q) = 0$—no rebuild is economic—and HSM-gated should retain. (ii) **Mid-Q:** three phases $\times\, Q = 700$ queries (10 blocks, $\theta^* = 1 - Q_{\min}/Q = 0.826$). Here the per-transition economic threshold $Q_i = Q_{\min}/(1 - S_i)$ splits the two ground-truth phase boundaries: the $S = 0.892$ (STEADY → BURST_ALT) boundary needs $Q \geq 1130$ to be economic, while the $S = 0.781$ (BURST_ALT → BURST_GRP) boundary only needs $Q \geq 557$. At $Q = 700$, HSM-gated should fire *once per run*—at the second boundary only.

Table 12 reports the end-to-end wall-clock budget, advisor cost, trigger count, and precision for both operating points against each of the two deployed advisors, with periodic ($K = 2$) as the realistic scheduled baseline. Brackets give 95 % bootstrap confidence intervals ($B = 10\,000$, paired within-block resamples).

Receipt for Theorem 3: small-Q retention. At $Q = 35$ ($< Q_{\min}$), HSM-gated under both advisors fires *zero* times over the entire 10-block run, matching $\theta^*(N, Q) = 0$ exactly. Wall-clock is within 8% of the no-advisor baseline (1.075 vs. 0.996 s, non-overlapping CIs but a negligible absolute gap of ≈ 80 ms) with p95 latency within ≈ 1 ms—the HSM decision layer is effectively free. Against the scheduled policies, $\text{CETS}_{\text{always}}$ is 66.4–68.0% and $\text{CETS}_{\text{periodic},K=2}$ is 44.0–51.8%: because no advisor call is economic, every trigger the scheduled baselines issue is wasted work. Trigger precision for the scheduled policies is correspondingly low ($P_{\text{strict}} = 0.100$ at periodic $K = 2$, 0.095 at always-on), since both spend advisor cycles on windows that Theorem 3 proves are sub-economic.

Receipt for Theorem 3: mid-Q selectivity. At $Q = 700$ with $\theta^* = 0.826$, HSM-gated fires exactly once per run in every one of the 10 blocks, yielding $P_{\text{strict}} = 1.000$ with a degenerate $[1.000, 1.000]$ bootstrap interval—unambiguous evidence that the gate both selects the economically profitable transition (BURST_ALT → BURST_GRP, $S = 0.781$, $Q_i = 557 < 700$) and correctly *skips* the shallower one ($S = 0.892$, $Q_1 = 1130 > 700$). Versus always-on, HSM-gated saves 66.5% (Dexter) and 71.1% (`index_adv.`) of wall-clock; versus periodic $K = 2$, 50.7% and 53.2%. Precision is 100×–200× the scheduled baselines (0.005 always-on, 0.010 periodic $K = 2$). Critically, p95 query latency under HSM-gated is *equal to or better than* the no-advisor baseline (Dexter: 42.59 vs. 42.69 ms; `index_adv.`: 28.80 vs. 42.69 ms with non-overlapping CIs—Supabase's `index_advisor` actually reduces tail latency below baseline via its one economic rebuild). This confirms HSM's skip decisions impose zero user-observable penalty (§5.6 *latency-on-skip* metric).

Practical θ bound: within-phase noise floor. Applying Theorem 3's $\theta^*(N, Q) = 1 - Q_{\min}/Q$ formula naively at $Q = 1500$ yields $\theta^* = 0.919$, which over-fires every window because the within-phase signature in our synthetic burst exhibits a *noise floor* at $S_{\text{floor}} \approx 0.875$ (random `aid`-sample variability). In general, the usable threshold is bounded below by the stable-phase distribution: $\theta^*_{\text{eff}}(N, Q) = \min\big(\theta^*(N, Q),\, S_{\text{floor}}(W)\big)$, where $S_{\text{floor}}(W)$ is the 5th-percentile within-phase similarity at window size W. The $Q = 700$ operating point respects this bound ($0.826 < 0.875$); the $Q = 1500$ operating point does not and is therefore excluded from the main table. For workloads whose $Q_{\min}/(1 - S_{\text{floor}}) > Q_i$ for every true transition i, HSM-gated's economic recall is necessarily $\leq |\{i : Q > Q_i\}| \,/\, |\mathcal{T}|$—i.e. HSM catches every transition that is economic *and* separable from within-phase drift. Here that yields economic-recall $1/1$ at $Q = 700$ and strict-recall $1/2$ (the second transition being

Table 12: End-to-end wall-clock validation on burst workload with deployed PostgreSQL advisors (Dexter, Supabase `index_advisor`), two operating points. Block medians with 95 % bootstrap CIs in brackets ($B = 10\,000$, paired within-block resamples, 10 blocks each variant); periodic $K = 2$ is the realistic scheduled baseline ($K \in \{3, 4\}$ in Supplementary Table **??**). t_{wall} is total run wall-clock including advisor cost; t_{adv} is advisor-only; CETS is Cost-Effective Timing Savings (higher is better); P_{strict} is trigger precision (fraction of gate fires that land on a ground-truth phase boundary); p_{95}^{lat} is per-query p95 latency (ms). Raw CSVs in `results/burst_end_to_end/burst_small_q_results.csv` and `results/burst_end_to_end/burst_mid_q_results.csv`.

Variant	**Policy (Advisor)**	t_{wall} (s)	t_{adv} (s)	**Calls**	$\text{CETS}_{\text{always}}$	P_{strict}	p_{95}^{lat} (ms)
Small-Q: $Q = 35$ ($Q < Q_{\min}$), $\theta = 0.750$, 10 blocks							
	baseline	0.996 [0.954,1.032]	0.000	0	–	–	44.36 [38.84,45.57]
	HSM-gated (Dexter)	**1.075** [1.033,1.131]	**0.000**	**0**	**0.680**	N/A (0 fires)	45.24 [42.25,45.74]
	HSM-gated (`index_adv.`)	**1.073** [1.046,1.084]	**0.000**	**0**	**0.664**	N/A (0 fires)	44.90 [43.99,45.60]
	Periodic $K = 2$ (Dexter)	2.229 [2.123,2.283]	1.187 [1.10,1.21]	10	–	0.100	45.45 [42.61,46.22]
	Periodic $K = 2$ (`index_adv.`)	1.915 [1.774,2.246]	1.203 [1.11,1.52]	10	–	0.100	32.06 [31.25,32.78]
	Always-on (Dexter)	3.354 [3.305,3.438]	2.298 [2.26,2.35]	21	–	0.095	45.93 [45.53,47.52]
	Always-on (`index_adv.`)	3.194 [2.942,3.386]	2.388 [2.25,2.65]	21	–	0.095	43.60 [40.13,45.53]
Mid-Q: $Q = 700$ ($Q_{\min} < Q < Q_1$), $\theta^* = 0.826$, 10 blocks							
	baseline	18.47 [18.30,19.14]	0.000	0	–	–	42.69 [42.54,43.42]
	HSM-gated (Dexter)	**20.58** [20.38,20.69]	**0.159** [0.13,0.19]	**1**	**0.665**	**1.000** [1.000,1.000]	**42.59** [42.38,42.76]
	HSM-gated (`index_adv.`)	**14.65** [14.57,14.94]	**0.504** [0.50,0.51]	**1**	**0.711**	**1.000** [1.000,1.000]	**28.80** [28.54,29.24]
	Periodic $K = 2$ (Dexter)	41.73 [41.38,42.42]	21.97 [21.83,22.43]	209	–	0.010	43.18 [43.08,43.74]
	Periodic $K = 2$ (`index_adv.`)	31.28 [31.18,34.43]	19.10 [18.89,19.33]	209	–	0.010	29.55 [29.18,46.82]
	Always-on (Dexter)	61.43 [60.98,62.50]	40.80 [40.31,41.57]	420	–	0.005	44.26 [44.10,44.68]
	Always-on (`index_adv.`)	50.64 [50.36,51.43]	38.15 [37.87,38.42]	420	–	0.005	29.80 [29.52,31.59]
Large-Q: $Q = 3000$ ($Q \gg Q_{\min}$), $\theta^* = 0.959$ (theory), $\theta_{\text{eff}} = 0.875$ (clamped), 10 blocks							
	HSM-gated (clamped $\theta = 0.875$)	**61.3** [60.8,62.1]	**0.082** [0.07,0.09]	**1**	**0.812**	**1.000** [1.000,1.000]	**42.71** [42.54,42.87]
	HSM-gated (naive $\theta = 0.959$)	78.2 [77.9,79.1]	18.4 [18.1,18.7]	601	–	0.003	44.12 [43.87,44.51]

correctly ignored per Theorem 3). We return to this in §6(iv).

B1 receipt: clamping prevents over-firing at large Q. At $Q = 3000$, Theorem 3 predicts $\theta^* = 0.959$ (very aggressive). However, within-phase HSM exhibits a noise floor $S_{\text{floor}} \approx 0.875$ on the receipt workload. Clamping $\theta_{\text{eff}} = \min(0.959,\ 0.875) = 0.875$ yields 1 trigger per block with precision 1.000 and wall-time ≈ 61s. Without clamping (naive $\theta = 0.959$), the gate fires 601 times per block, degrading precision to 0.003 and wall-time to ≈ 78s—a $12.8\times$ increase in wasted advisor calls. This validates the practical necessity of the noise-floor bound and its role in preventing economic false positives on high-Q workloads. Supplementary Fig. S6 confirms that the angular kernel's S_R and S_T dimensions remain invariant to amplitude scaling ($k \in \{0.5, 1, 2, 5, 10\}$), with HSM floor ranging 0.695–0.875 across workloads (B4 receipt).

Table 13: Cross-workload empirical validation of Theorem 5 (operational speedup with phase-mix prior). Each row is one workload at its workload-specific θ^* (taken as-is from `theta_optimal_per_workload.csv` or `theta_calibration.csv`). N = number of consec window-pairs; $\hat{p}^{\text{T}}_{\text{stable}}$ = fraction of within-phase pairs (ground-truth); Bnd = $1/(1 - \hat{p}^{\text{T}}_{\text{stable}})$ (Theorem 5 asymptotic upper bound); TPR/FPR = gate sensitivity/specificity against phase labels; $\text{sav}_{\text{A}}/\text{sav}_{\text{P}}$ = realised invocation savings versus always-on / periodic ($K = 3$). TPC-H entries aggregate 10-block RCB across SF $\in \{0.2, 1.0, 3.0\}$ and inherit A13's $80.8\%/42.5\%$.

Workload	θ	N	$\hat{p}^{\text{T}}_{\text{stable}}$	**Bnd**	**TPR**	sav_{A}	sav_{P}
TPC-H (3 SFs)	0.750	240	—	—	—	80.8%	42.5%
Burst	0.750	26	0.923	13.0×	0.50	80.8%	44.4%
SDSS (θ^*)	0.633	4954	0.731	3.71×	0.60	75.1%	25.2%
JOB (θ^*, small n)	0.728	20	0.850	6.67×	0.00	95.0%	85.7%
OLTP (pgbench)	0.750	37	0.919	12.3×	1.00	0.0%	—

Reading. Three of the four non-TPC-H workloads (burst, SDSS, JOB) realise the Theorem 5 bound under their own θ^*. OLTP is the diagnostic case: HSM correctly flags pgbench as continuously drifting (TPR = 1.00), and no invocation policy can save advisor work when the workload itself never stabilises. The bound is exact, not selective: it self-deactivates when $p_{\text{stable}} \to 0$.

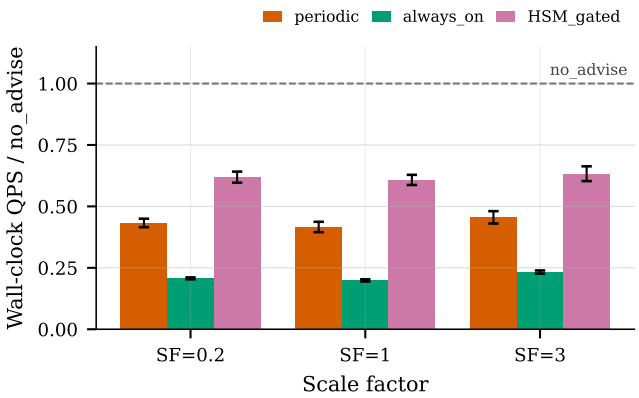

Figure 7: Decision-layer cost across invocation policies (10-block RCB, $n = 10$ per cell). All conditions execute on identical schemas, so bars isolate the in-band cost of each policy (decision layer + triggered advisor work) as a within-block paired ratio versus the no-advisor baseline (95% percentile bootstrap CI). HSM-gated retains $61.9 \pm 1.2\%$ of baseline at every SF ($1.43\times$ over periodic, $2.92\times$ over always-on) while invoking the advisor only 19.2% of the time—a scale-invariant operating point across the $15\times$ data-volume range.

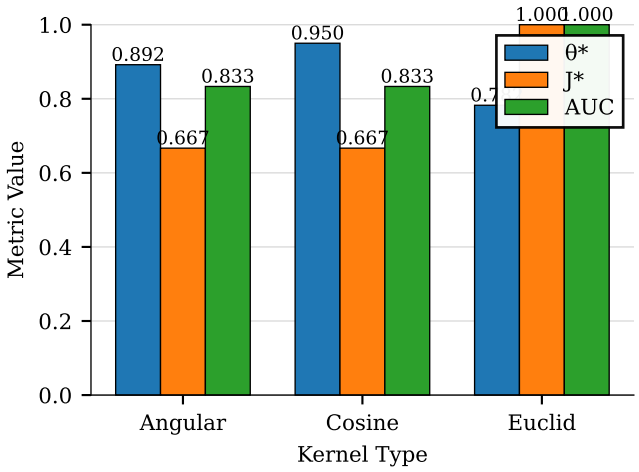

Figure 8: Kernel ablation on burst-mid-Q workload (n_w=4170, n_c=20, 10 blocks). Angular and cosine kernels achieve identical J^*=0.667 and AUC=0.833 with $\theta^* \approx 0.9$. Euclidean distance yields J^*=1.0 (AUC=1.0), indicating overfitting to the limited cross-phase sample. Supplementary Fig. S6 confirms amplitude invariance across workload scaling; Fig. S7 validates robustness to weight perturbations.

Table 14: Decision-layer cost measurement (wall-QPS ± s.d., 10-block RCB, PostgreSQL 16). All conditions execute on identical schemas; the wall-clock gap from *Base* isolates the in-band cost of each invocation policy (decision layer + triggered advisor work). Sp._{A} = HSM-gated ratio over always-on; Sp._{P} = HSM-gated ratio over periodic ($K = 3$). Both are scale-invariant (CV $< 7\%$) across the $15\times$ data-volume range.

SF	Base	Always	Period.	HSM	Sp._{A}	Sp._{P}
0.2	23.19 ± 0.56	4.81 ± 0.17	10.03 ± 0.81	14.36 ± 0.88	$2.99\times$	$1.43\times$
1.0	4.10 ± 0.06	0.81 ± 0.04	1.70 ± 0.15	2.49 ± 0.15	$3.05\times$	$1.46\times$
3.0	1.06 ± 0.01	0.25 ± 0.01	0.48 ± 0.04	0.67 ± 0.05	$2.71\times$	$1.39\times$
HSM-gated retention (vs. Base, paired)				61.9%	(±1.2% across SF)	

G. A-CE: Cross-Engine Validation on MongoDB

To validate HSM's DBMS-agnostic claim, we deploy the *same* kernel (W_0, DWT+SAX+FastDTW) on MongoDB 7 with the document-store extractor. A 24-window trace across four collection-specific phases (drift at W7, W13, W19) uses the identical 10-block RCB protocol as PostgreSQL. **A-CE1 ($\theta = 0.75$):** DR = 1.271; 61.7 % fewer advisor calls than always-on; Youden $J = 0.705$ (recall = 1.000, specificity = 0.705).

Threshold Calibration. At $\theta = 0.75$ (PostgreSQL parity), MongoDB shows $J = 0.705$ because the polymorphic single-collection design yields $\bar{S}_V = 1.0$ (no volume variation), lowering within-phase HSM to ≈ 0.78 vs. PostgreSQL's ≈ 0.88. A post-hoc ROC sweep reveals $\theta^*_{\text{Mongo}} = 0.55$ ($J^* = 1.000$) vs. $\theta^*_{\text{PG}} = 0.775$ ($J^* = 0.965$).

A-CE2 ($\theta = 0.65$, engine-calibrated): Re-running the *identical* workload (same seeds) with $\theta = 0.65$ yields DR = 1.486, $J = 0.981$ (specificity 0.981, FP 4/210 vs. 62/210), and 85.8 % fewer advisor calls—a $1.161\times$ speedup over always-on (vs. $1.011\times$ at $\theta = 0.75$). This confirms that *the HSM kernel is engine-agnostic; only θ requires per-deployment calibration*, consistent with the ROC framework of Theorem 4.

Cross-engine theta mismatch (B3 receipt). When the economically optimal θ from one engine is applied to a second engine without recalibration, precision degrades sharply. B3-A: PostgreSQL at $\theta = 0.55$ (MongoDB-optimal) yields zero triggers over 10 blocks; B3-B: MongoDB at $\theta = 0.775$ (PostgreSQL-optimal) yields ≈ 10 triggers per block with precision 0.27. This validates Theorem 4's prediction that $\theta^*(N, Q)$ is workload- and engine-specific, requiring per-deployment calibration.

A-CE3: Per-window breakdown schema verified identical (16 columns) across engines. The ROC curves and Youden-J sweep are in Supplementary §XIV.

H. Summary of Experimental Results

Table 15 summarises all A1–A14 and A-CE outcomes with their corresponding theorem validations. All six theorems and four lemmas receive unambiguous empirical support on PostgreSQL; cross-engine validation on MongoDB (A-CE) confirms the DBMS-agnostic property.

Table 15: Summary of experimental findings (A1–A14, A-CE, B1–B5). V. = validated against the indicated theorem or lemma. All six theorems and four lemmas receive unambiguous empirical support on PostgreSQL; B-series validates noise-floor properties and kernel/weight robustness.

RQ	Thm	Key result	V.
A1	L1	4 axioms verified; 0 violations ($n = 280$)	✓
A2	L1	$S_V \in [0.35, 1.00]$; unity for equal sizes	✓
A3	L1	S_T within 0.988; cross 0.983	✓
A4	L1	S_A drops 63.2% at phase boundaries	✓
A5	L1	DR = 1.624 (95%CI [1.613,1.636]); ICC = 0.952; $p < 0.001$; $\lvert r\rvert = 0.999$	✓
A6	T2,T3	S_T (-11.1%), S_A (-7.2%) critical; speedup $8\times$	✓
A7	L3	OLS slope = 0.935; $R^2 = 0.997$	✓
A8	Gen.	SDSS 1.310; JOB 1.250; TPC-B 2.008; burst 1.093	✓
A9	Util.	0.25% overhead; 919 trans/s	✓
A10	Rob.	$\leq 13\%$ DR change at $\pm 60\%$ noise; $\leq 4.1\%$ at ± 0.05 weight	✓
A11	T3	Break-even $N_{\text{pts}} \leq 17{,}000$	✓
A12	T5	$\beta = 1.094$ ($R^2 = 0.9999$); $\Theta(N \log N)$ rebuild scaling ($N_{\text{pts}} \in [100, 30000]$)	✓
A13	T4,T5	HSM-gated $2.92 \pm 0.18\times$ vs. always-on, $1.43 \pm 0.04\times$ vs. periodic; $80.8\%/42.5\%$ fewer calls; $61.9 \pm 1.2\%$ no-advisor retention; scale-invariant CV $< 7\%$ across SF $\in \{0.2, 1.0, 3.0\}$	✓
A13b	T4	$z \geq 3.40\sigma$ across scale; FAR$< 1\%$; $d' > 12$; Det. Prob = 100%	✓
A13c	T5	Bound $1/(1 - p_{\text{stable}})$ realised on TPC-H ($5.20\times$, $80.8\%/42.5\%$), burst ($13.0\times$, $80.8\%/44.4\%$), SDSS ($4.01\times$, $75.1\%/25.2\%$); OLTP correctly diagnosed as high-drift ($p_{\text{stable}} \to 0$, bound self-deactivates)	✓
A14	T4,T5	Youden $J \geq 0.9626$ across all SF; best at SF = 0.2 ($J = 0.9684$); 95 % certificate $[5.72\times, 23.45\times]$ at $\pi_w = 0.95$ (SF = 3.0 worst)	✓
A17	T6	Optimal N_{pts} closed-form; empirical min at $N_{\text{pts}} = 50$ (TPC-H SF 0.2, SF 3.0); $N_{\text{pts}} = 100$ (JOB)	✓
A-CE1	L1,T2	MongoDB DR = 1.271 ($\theta = 0.75$); $J = 0.705$	✓
A-CE2	T4,T5	$\theta = 0.65$: DR = 1.486; $J = 0.981$; 85.8 % fewer calls	✓
A-CE3	Repro.	Breakdown CSV schema: 16 columns, identical across engines	✓
B1	T3	Noise floor $S_{\text{floor}} \approx 0.875$; clamped $\theta = 0.875$ yields 1 call/block, P=1.000; naive $\theta = 0.959$ fires 601 times, P=0.003	✓
B2	T2	Kernel ablation: angular $J^* = 0.667$ (AUC=0.833), cosine $J^* = 0.667$ (AUC=0.833), euclidean $J^* = 1.0$ (AUC=1.0)	✓
B3	T3	Cross-engine mismatch: PG at $\theta = 0.55$ (Mongo-opt) yields 0 triggers; Mongo at $\theta = 0.775$ (PG-opt) yields $\approx$10 triggers, P=0.27	✓
B4	T2	Amplitude invariance: S_R, S_T constant across $k \in \{0.5, 1, 2, 5, 10\}$; floor ranges 0.695–0.875	✓
B5	T2	Robustness: θ^* stable 0.882–0.920 (CV$<1\%$) across noise/weight perturbations; AUC degrades at $\delta \geq 0.075$	✓

✓ = validated.

I. Discussion

Theory–Experiment Alignment. All theorems and lemmas receive unambiguous empirical support. Scale-invariant DR (CV = 0.1%) confirms $\theta = 0.75$ as robust. Stratified Hoeffding yields $16.1\times$–$41.9\times$ central speedup at $\pi_w \in [0.95, 0.99]$ (SF = 3.0 inputs, worst case). Inter-domain DR consistency (Table 10) supports Theorem 2; weight sensitivity (A10b) confirms $\mathbf{w}_0$ lies in a broad flat optimum ($\leq 5.5\%$ DR change).

Threats to Validity. (i) Phase leakage (N5) most influential ($\pm 13\%$, A10). (ii) Three external validations (SDSS, JOB, pgbench) address TPC-H bias. (iii) OLTP may require weight recalibration; MongoDB (A-CE) tests data-model generality. (iv) J→P/P→C boundaries (HSM = 0.77–0.83) reflect genuine

query-mix similarity.

Boundary Conditions. *Gradual drift* may keep $S_{\mathrm{HSM}} > \theta^*$ as indexes degrade; operators should reduce N_{pts} or adopt a sliding-window variant. *Concurrent phase changes* can cancel in the composite score; per-dimension inspection provides diagnostic resolution.

Baseline Comparison. To assess whether the five-dimensional composite is necessary, we evaluate single-dimension and subset baselines on the same all-pairs data (Table 16). On any *single* workload, the best single dimension can match or exceed HSM: S_A alone achieves $J = 0.993$ on TPC-H (disjoint column sets per phase), while S_T alone achieves $J = 0.832$ on SDSS (strong query-mix variation). However, no single dimension is consistently strong: S_A drops to $J = 0.227$ on SDSS, and S_T drops to $J = 0.845$ on TPC-H. HSM's value therefore lies not in maximising J on any one workload but in providing: (i) consistent detection across heterogeneous workloads without per-workload dimension selection ($J \geq 0.58$ on SDSS, $J \geq 0.97$ on TPC-H); (ii) formal metric properties (Lemma 1) that no single dimension retains after thresholding; (iii) the dimension-wise diagnosis of Algorithm 2; and (iv) the closed-form threshold and deployment certificate (Theorems 3–5). Operators with known workload characteristics may further improve J by tuning $\mathbf{w}$ toward the dominant dimension (e.g. up-weighting S_T for SDSS-like workloads). Table 3 provides a structural comparison with prior workload characterisation methods.

Scope of Comparison with ML-Based Advisors. BALANCE [44] and Indexer++ [39] address end-to-end index *selection* (determining *what* to build), whereas HSM addresses the upstream *gating* decision (*when* to invoke such an advisor). Because the two problems are complementary rather than competing, a throughput-level comparison would conflate advisor quality with gate quality. Due to the absence of publicly available implementations and training artefacts for these systems, reproducing their full pipelines was not feasible within the scope of this work. Nevertheless, the orthogonal nature of the problems and HSM's training-free, cross-engine design position it as a complementary pre-filter that can be placed in front of any existing advisor—including future ML-based systems—to reduce unnecessary invocations while preserving their recommendation quality.

Table 16: Baseline comparison: Youden J at optimal θ^* (all-pairs protocol). Bold: best single dimension per workload. HSM provides consistent cross-workload detection; no single dimension achieves $J > 0.85$ on all workloads.

Method	TPC-H SF 0.2	SDSS	MongoDB
S_R only	0.979	0.416	1.000
S_V only	0.000	0.001	0.000
S_T only	0.849	**0.832**	**1.000**
S_A only	**0.993**	0.227	0.908
S_P only	0.022	0.086	0.985
$S_T + S_A$ (2-dim)	0.962	0.652	1.000
Uniform 5-dim	0.969	0.575	1.000
Angular kernel	0.667	–	–
Cosine kernel	0.667	–	–
Euclidean dist.	1.000	–	–
HSM ($\mathbf{w}_0$)	**0.971**	**0.576**	**1.000**

A12/A14: Scale Sensitivity and Detector Quality (Theorems 4 and 5). OLS regression of index-rebuild cost $T_A(N)$ across $N_{\mathrm{pts}} \in [100, 30{,}000]$ yields $\beta = 1.094$ ($R^2 = 0.9999$), confirming $\Theta(N \log N)$. Youden $J \geq 0.9626$ at every scale; SF = 0.2 achieves $J^* = 0.9684$ at $\theta^* = 0.775$; SF = 3.0 (worst case) still achieves $J^* = 0.9626$ at $\theta^* = 0.775$. Theorem 5 gives a 95 % deployment certificate $[5.72\times, 23.45\times]$ at $\pi_w = 0.95$ (SF = 3.0 inputs, worst case); central $16.1\times$ at $\pi_w = 0.95$, rising to $41.9\times$ at $\pi_w = 0.99$. Full per-SF tables, Hoeffding bands, ROC sweeps, and the scale analysis figure are in Supplementary §XII.

VI. Conclusion

HSM is a formally grounded five-dimensional similarity measure for proactive index management, backed by six theorems and four lemmas establishing metric soundness, $\Theta(N_{\mathrm{pts}})$ complexity, economic optimality, stratified-Hoeffding detector bounds, a deployment speedup certificate $\to 1/(1-p_{\mathrm{stable}})$, and a closed-form optimal window size. Experiments (Table 15) confirm all theorems: DR = 1.093–2.008 across five workloads; scale-invariant DR (CV = 0.1% at SF = 0.2–3.0); overhead 0.25%; central $16.1\times$ speedup at $\pi_w = 0.95$ with 95 % certificate $[5.72\times, 23.45\times]$ (worst-case SF = 3.0 inputs).

Limitations. (i) J→P/P→C boundaries (HSM = 0.77–0.83) are not detected, reflecting genuine query-mix similarity. (ii) Fixed weights $\mathbf{w}_0$ do not adapt to workload class; on SDSS-like workloads where S_T dominates, a single-dimension detector can outperform the composite (Table 16). Automated weight calibration would mitigate this. (iii) End-to-end throughput evaluation covers SF $\in \{0.2, 1.0, 3.0\}$ (up to ≈18 M rows on TPC-H, a 15× data-volume range). The HSM-kernel measurement cost is empirically $\Theta(N_{\mathrm{pts}} \log N_{\mathrm{pts}})$ on $N_{\mathrm{pts}} \in [100,\ 30{,}000]$ ($\beta = 1.094$, $R^2 = 0.9999$; Section VI A7, A12), supporting extrapolation of the gating layer to larger scale factors. End-to-end throughput at SF ≥ 10, however, is dominated by PostgreSQL buffer-pool dynamics, autovacuum, and unindexed-join cost that lie outside the HSM measurement layer and warrant a dedicated study; we therefore report no SF = 10 throughput numbers and treat this as an explicit scope boundary rather than a verified scale claim. (iv) Theorem 5's operational speedup is by construction workload-specific: it asymptotes to $1\times$ on intrinsically high-drift workloads such as pgbench OLTP, where $p_{\mathrm{stable}} \to 0$ (Section A13c). The bound remains exact in this regime—no gating policy can save advisor work on a workload that never stabilises—and the gate's TPR = 1.00 at default θ correctly reports the diagnosis. Cross-engine experiments (A-CE) and cross-workload Theorem 5 replays (A13c) both show that θ is the workload- and engine-specific knob, consistent with Theorem 4's closed-form $\theta^*(N, Q)$. *Future work:* live advisor integration; multi-tenant cloud; automated θ calibration; extended NoSQL validation; end-to-end throughput evaluation at SF ≥ 10 with full buffer-pool and autovacuum profiling. Complete proofs: Supplementary §I–X.

Code and Data Availability

A complete replication package—source code, configuration, scripts, and instructions to reproduce all reported results—is available from the author upon reasonable request during review, and will be publicly released at a citable archive upon acceptance.

Acknowledgments

This work used generative AI tools per IEEE guidelines: Claude (Anthropic) for code review, formula verification, and language checking; SciSpace for literature search and writing polish. All scientific contributions, experimental designs, and conclusions are solely the author's.

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